

Learning from Videos

Feb 19th, 2024

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Videos

Videos are all around us

Span an enormous space of spatial and temporal signals





Challenges in Videos: size of video

Image



video



Size of video >> size of image

Computational constrains → short, low-res clips

3 x H x W



4D tensor:
T x 3 x H x W

time



~30 frames per second (fps)

Uncompressed size (3 bytes per pixel):
SD (640 x 480): **~1.5 GB per minute**
HD (1920 x 1080): **~10 GB per minute**

Reduce spatial and temporal resolution



5fps, half the spatial resolution

Slide inspiration: Justin Johnson, EECS 498-007

Challenges in Videos: size of video

Computational constrains → short, low-res clips



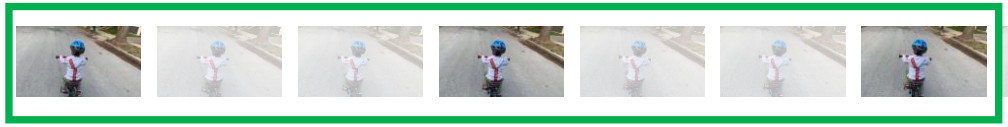
Input video

Walking
Running
Cycling
Jumping
.
.

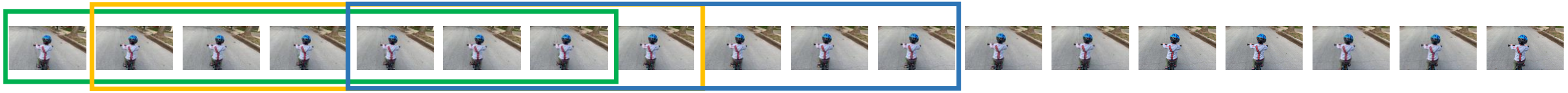
Original video(long, high FPS)



Training: Short, low FPS



Test time: inference on different short clips, average predictions



Slide inspiration: Justin Johnson, EECS 498-007

Challenges in Videos: Videos Datasets

space of video >> space of image → lots of training data

"ImageNet"-equivalent dataset for videos?

Massive human labelling efforts



UCF101

YouTube videos

13320 videos, 101 action categories

Kinetics

Kinetics

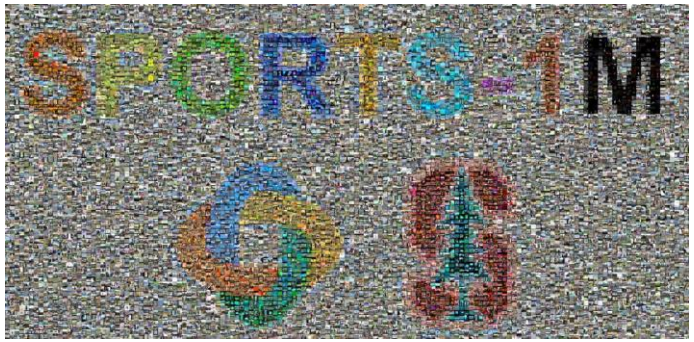
YouTube videos

650,000 video clips, 600 human action classes

YouTube | 8M

YouTube-8M

8M video clips, Machine-generated annotations from 3,862 classes



Sports-1M

YouTube videos

1,133,157 videos, 487 sports labels

Today

Deep Learning-based Models for Videos

- How to reduce computation cost without sacrificing accuracy?
- What architecture to best capture temporal patterns?

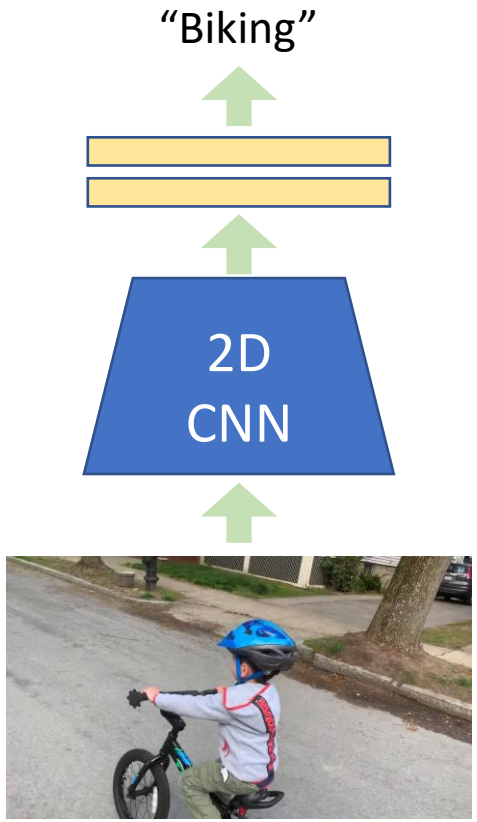
Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Self-Supervision in Videos

- Which types of pretext tasks can we define to capture temporal information?
- Learning from a single video and neural video represenation

Models for Videos: Single-Frame Baseline

- Train 2D CNN to classify video frames independently



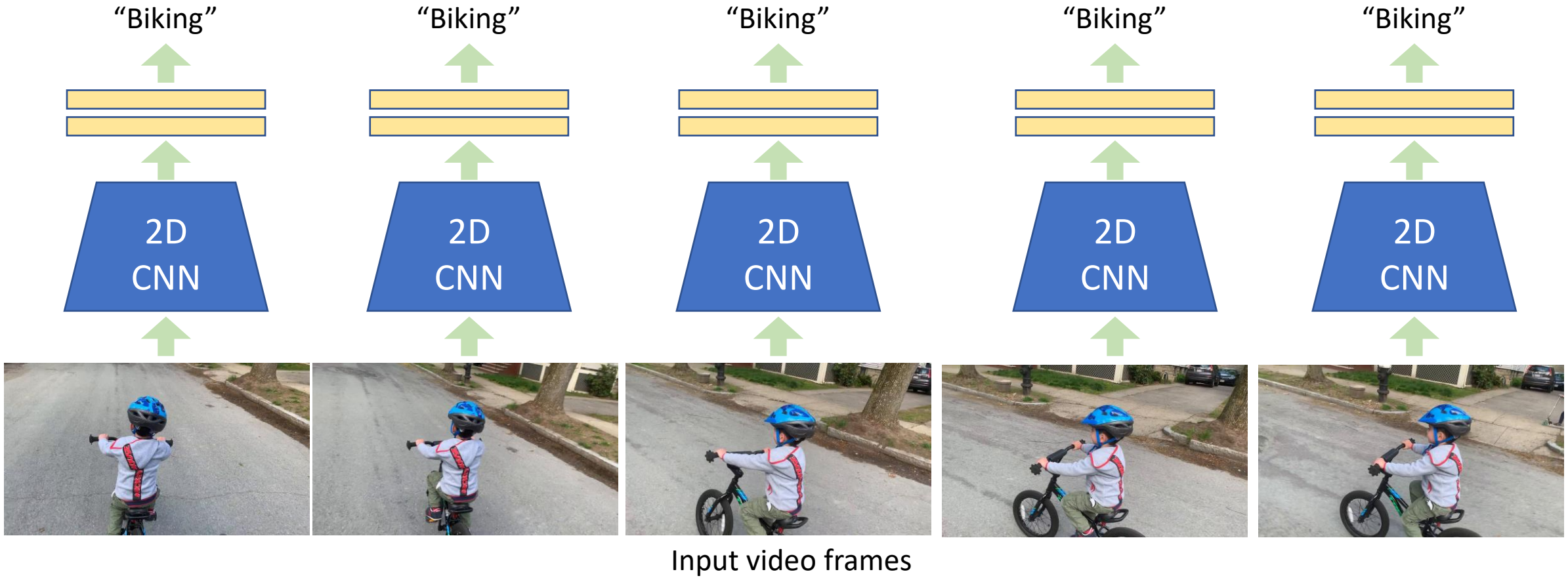
Input video frame

Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Single-Frame Baseline

- Train 2D CNN to classify video frames independently
- Average predicted probs at test-time

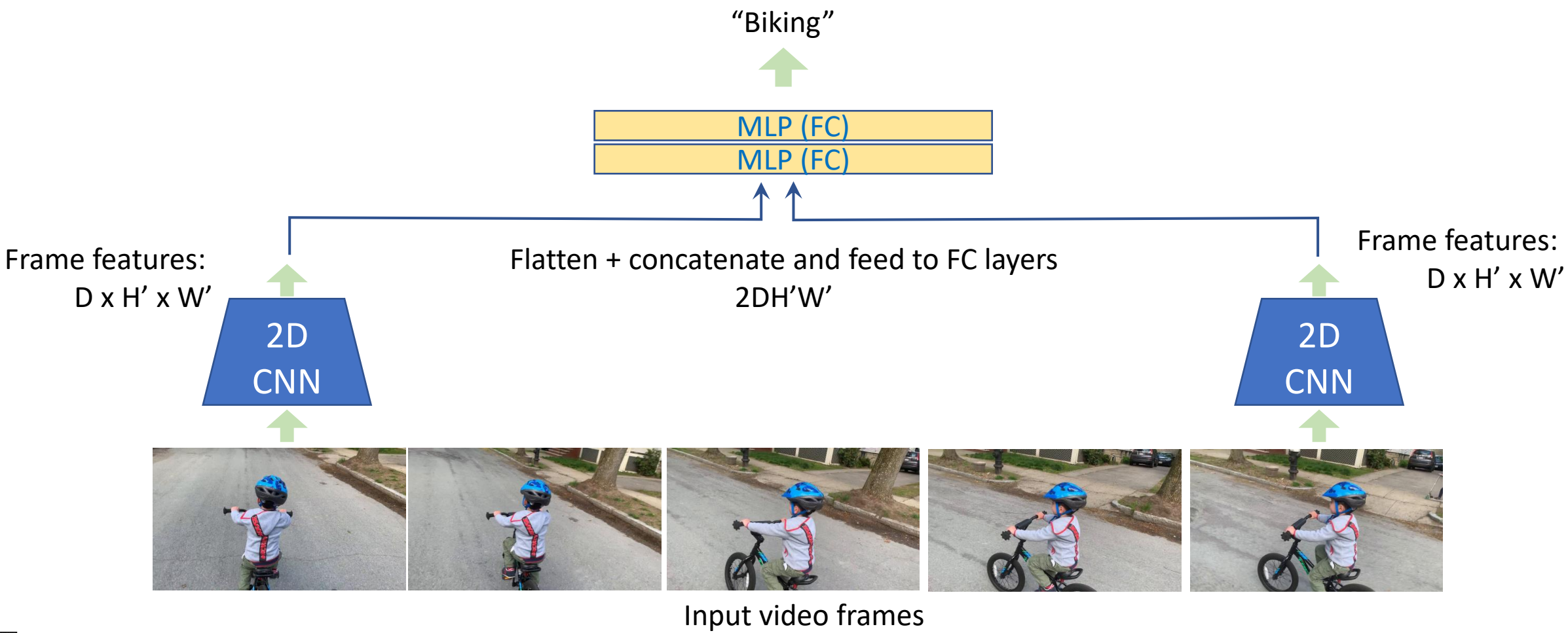
Often a surprisingly strong baseline!



Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Late Fusion

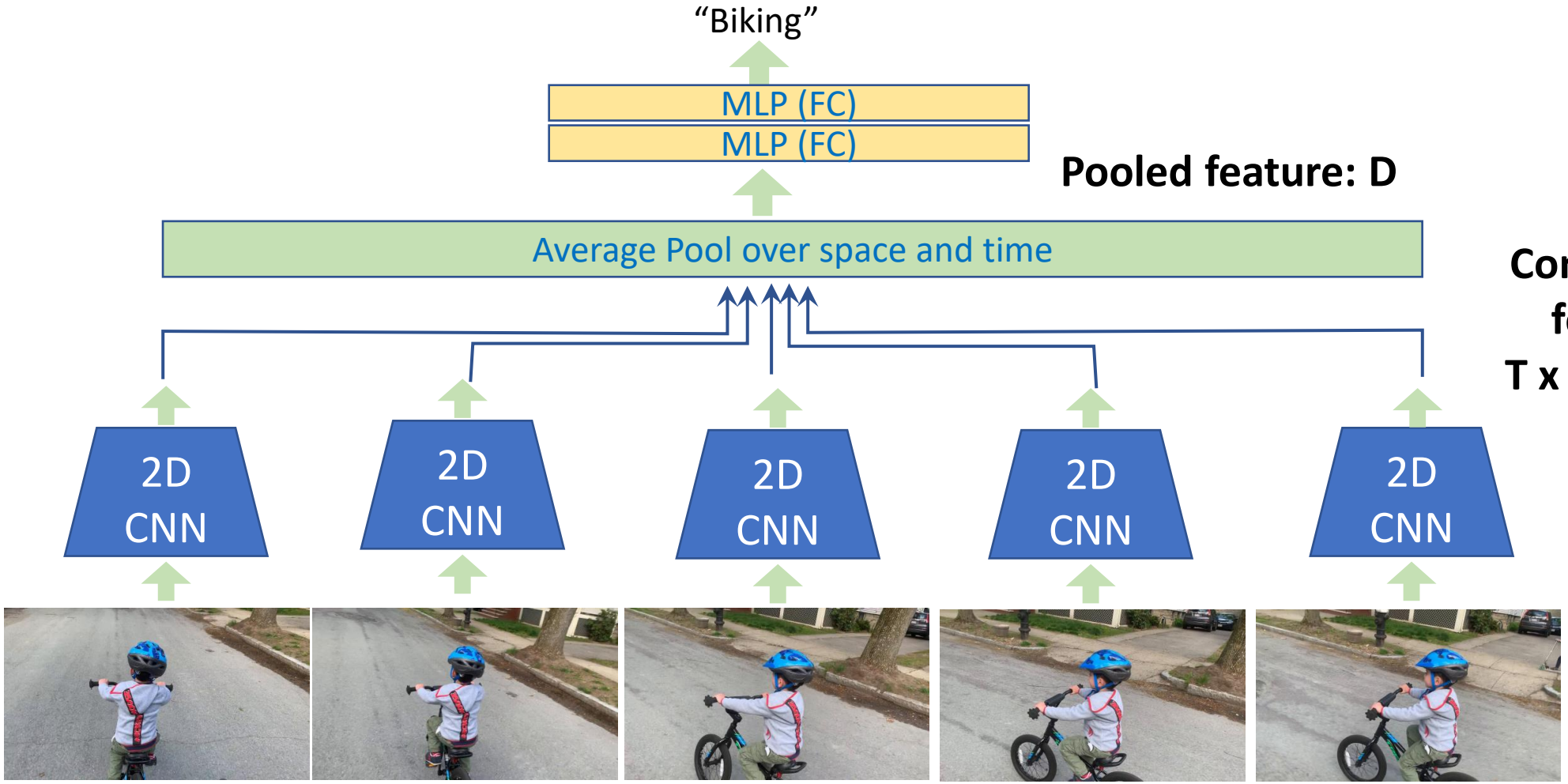
- Learn features for each frame using a 2D CNN, concatenate feature, and fuse



Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Late Fusion w/ pooling

Learn features for each frame, apply spatial-temporal average pool, and then fuse



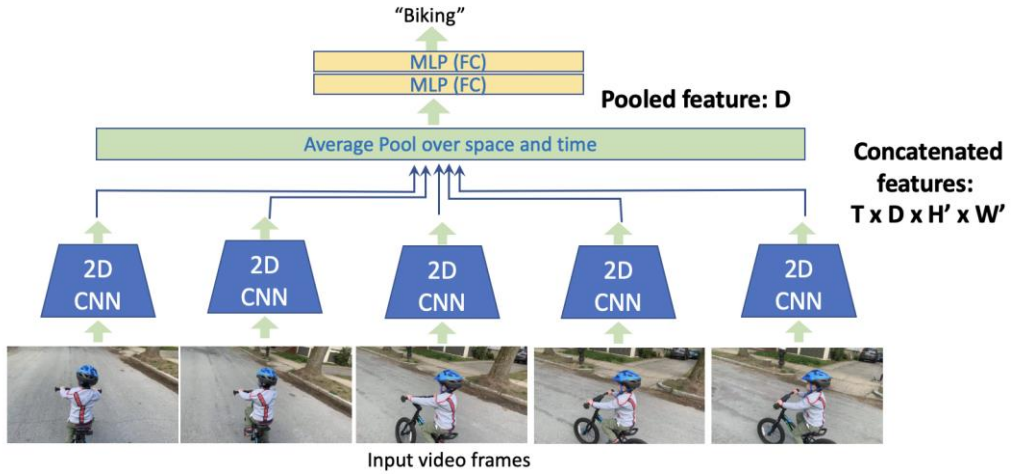
Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Late Fusion w/ pooling

Learn features for each frame, apply spatial-temporal average pool, and then fuse

Pros: allow the network to learn global motion characteristics by comparing outputs of both towers

Cons: late fusion is late…
hard to represent low level motion between frames



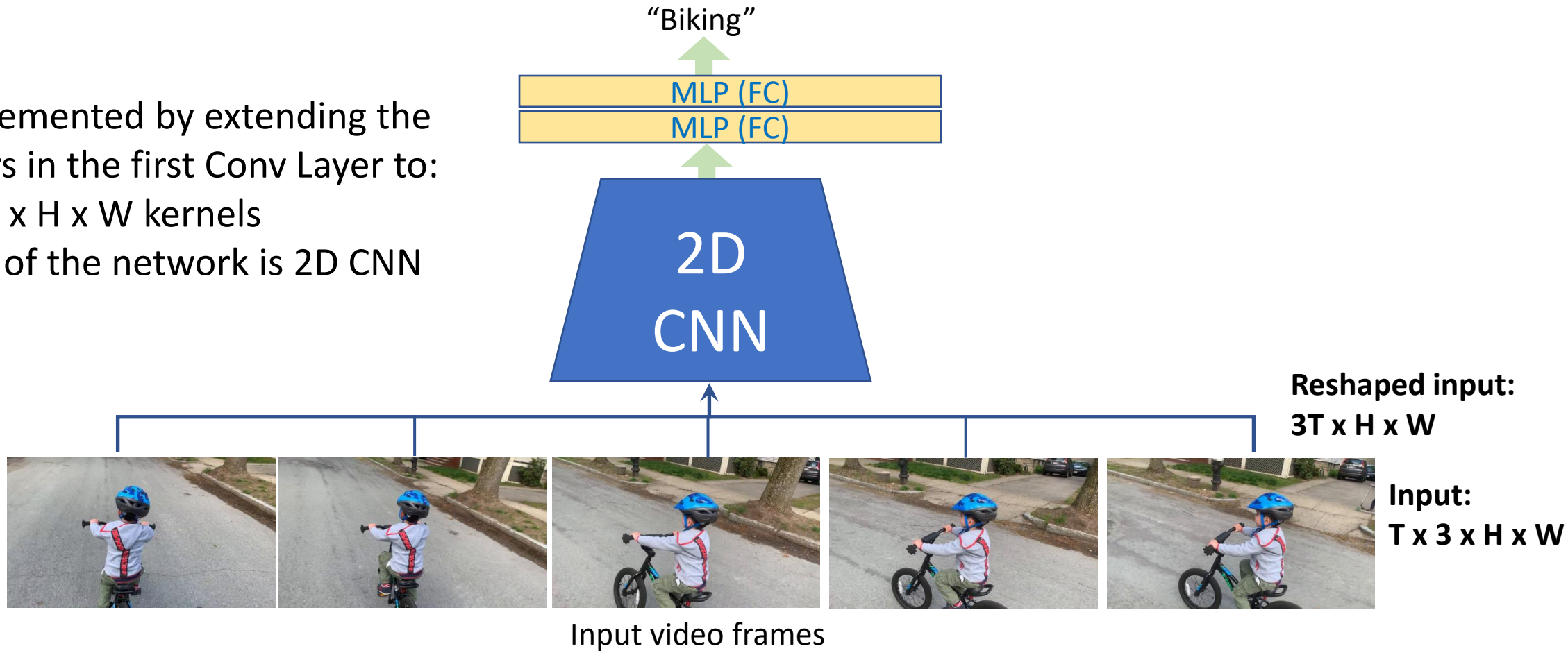


Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Early Fusion

- Combines temporal information immediately on the pixel level
- Treat time as another “channel” dimension

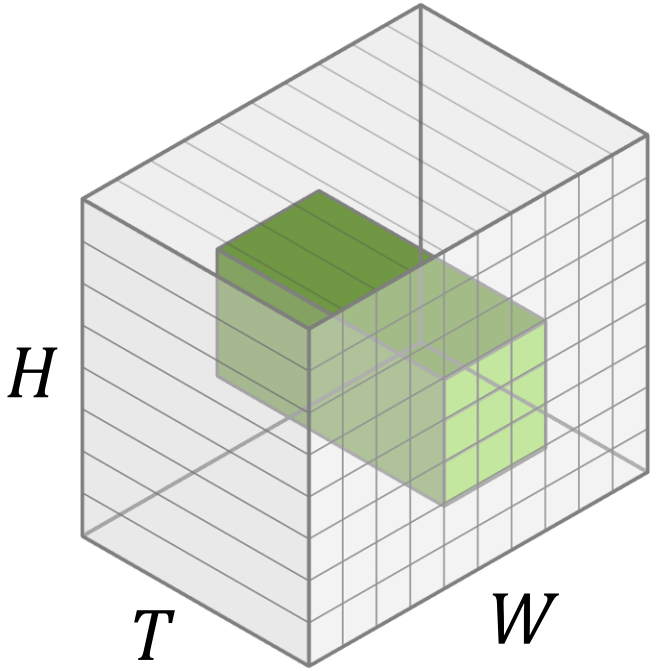
Implemented by extending the filters in the first Conv Layer to:
T x 3 x H x W kernels
Rest of the network is 2D CNN



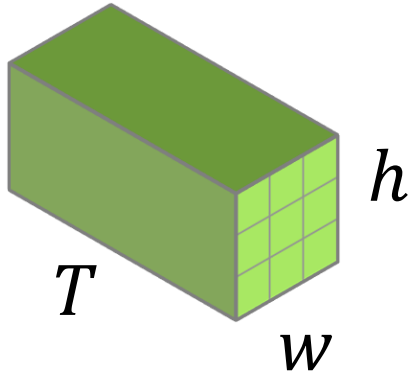
Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Early Fusion

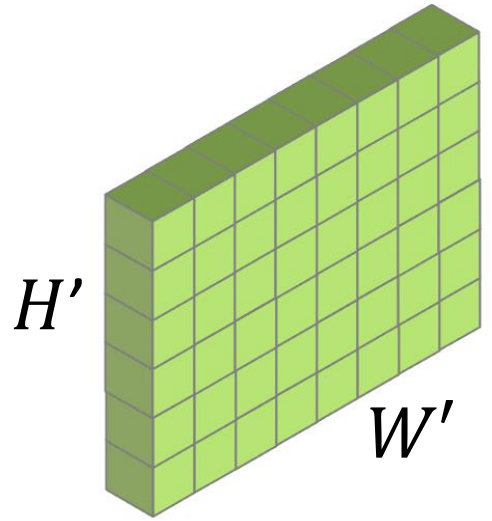
Extending the filters in the first Conv Layer to: T x 3 x H x W kernel



Input: T x 3 x H x W



Weights: C x T x 3 x h x w



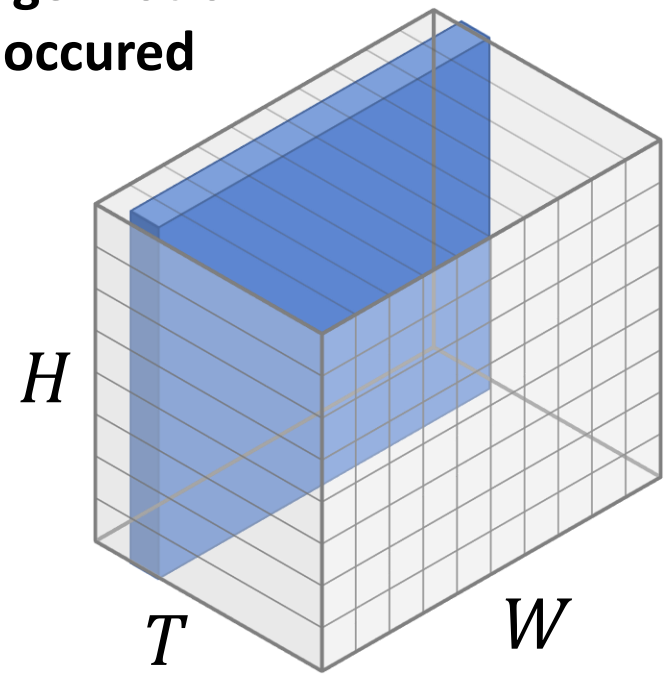
Output: C x H' x W'

Models for Videos: Early Fusion

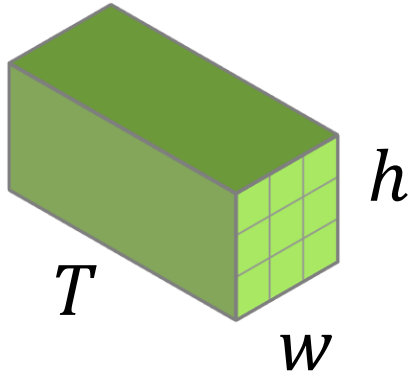
Extending the filters in the first Conv Layer to: T x 3 x H x W kernel

- Not temporal shift invariance; specific filter is learned to each time step

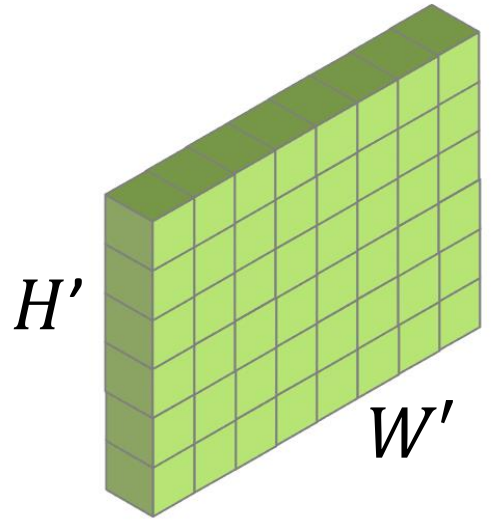
Large motion occured



Input: T x 3 x H x W



Weights: C x T x 3 x h x w



Output: C x H' x W'

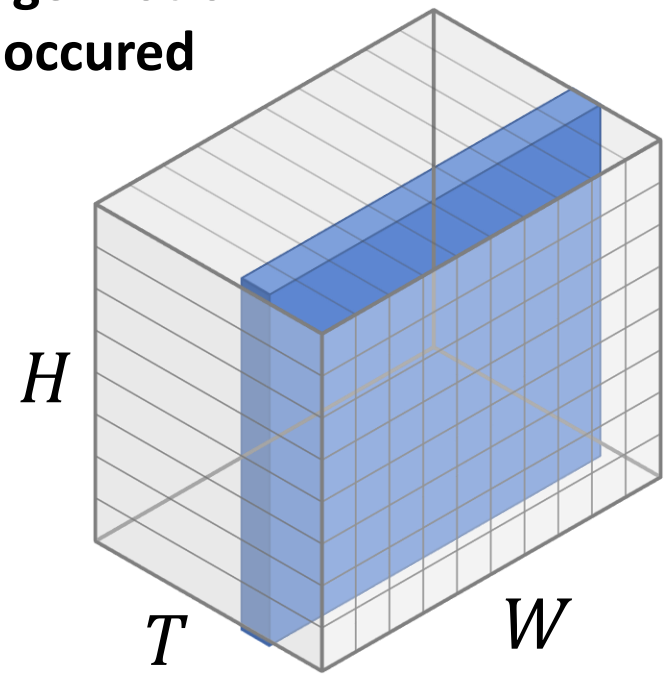
Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

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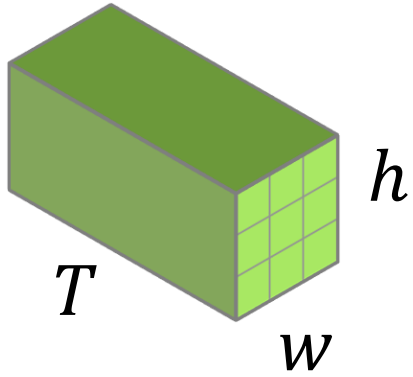
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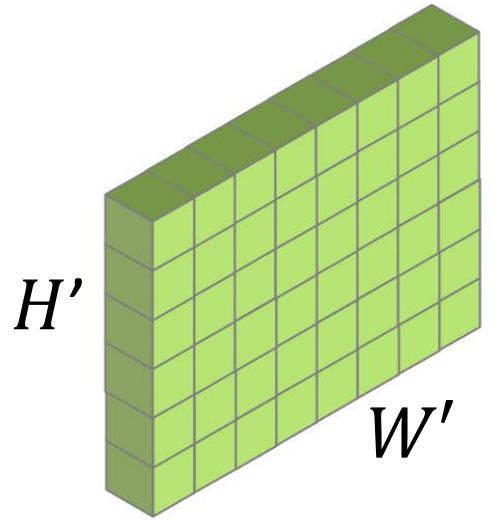
Large motion occured



Input: T x 3 x H x W



Weights: C x T x 3 x h x w



Output: C x H' x W'

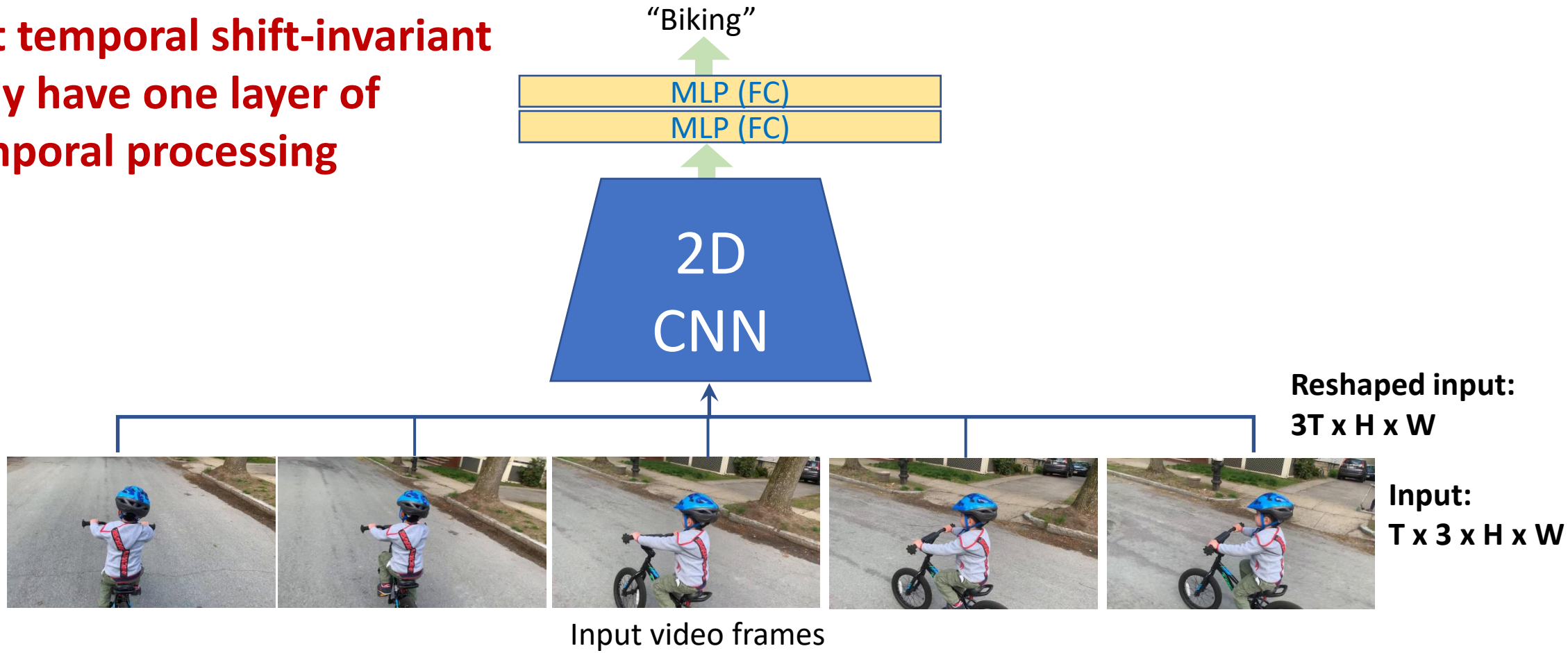
Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Early Fusion

Pros: Allow the network to learn local motion characteristics

Cons:

- **Not temporal shift-invariant**
- **Only have one layer of temporal processing**

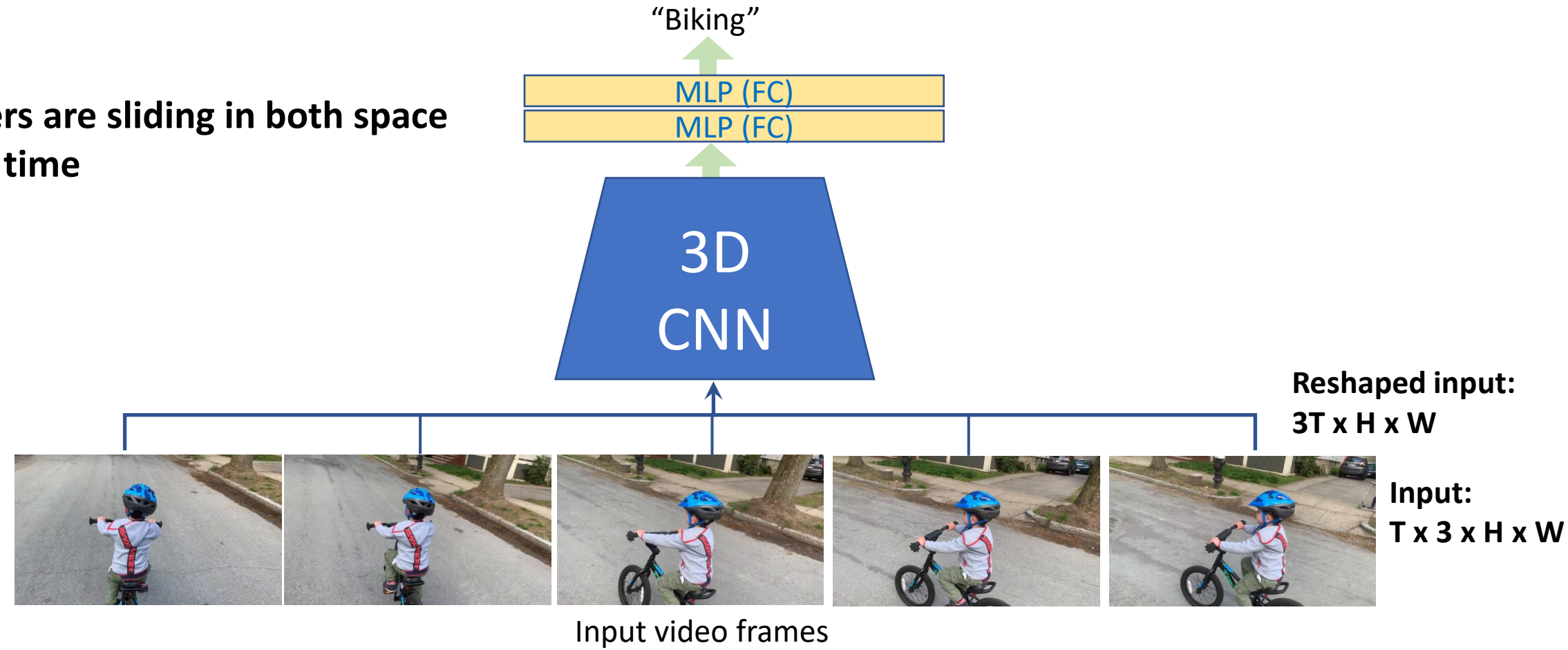


Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Slow Fusion a.k.a 3D Convs

- Extend 2D Convs and pooling to 3D to slowly fuse temporal information throughout the model

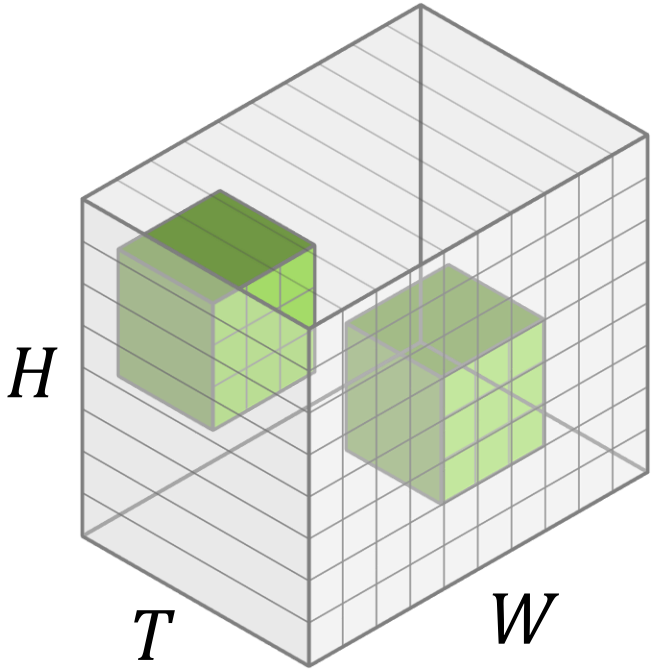
Filters are sliding in both space and time



Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

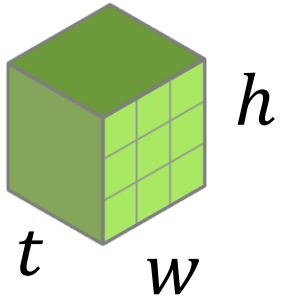
Models for Videos: Slow Fusion a.k.a 3D Convs

- Extend 2D Convs and pooling to 3D to slowly fuse temporal information throughout the model
- **Slide the kernels in both space and time**

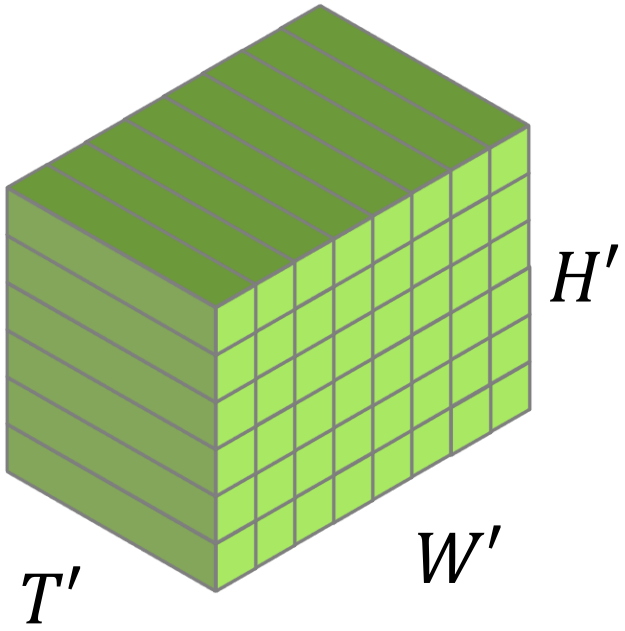


Input: T x 3 x H x W

- **Temporal shift-invariant!**



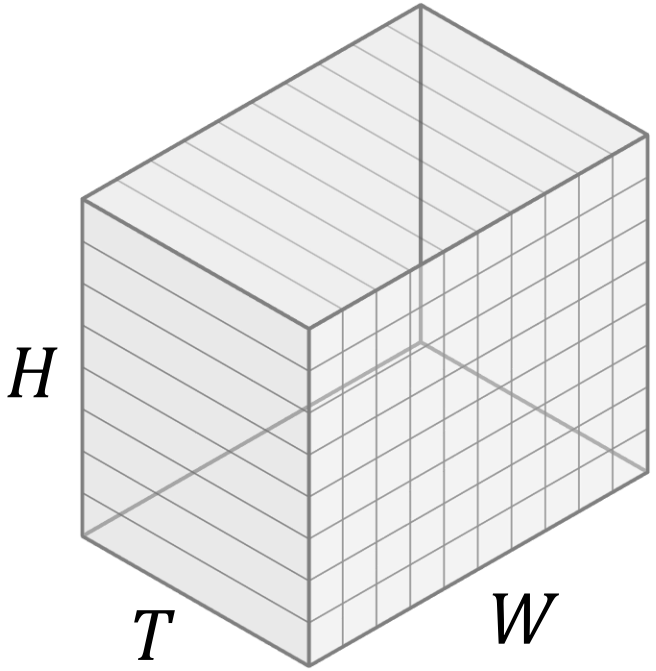
Weights: C x t x 3 x h x w



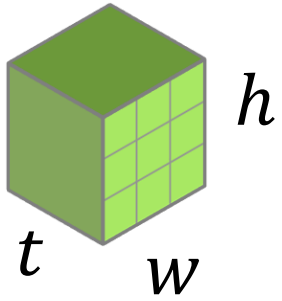
Output: C x T' x H' x W'

Models for Videos: Slow Fusion a.k.a 3D Convs

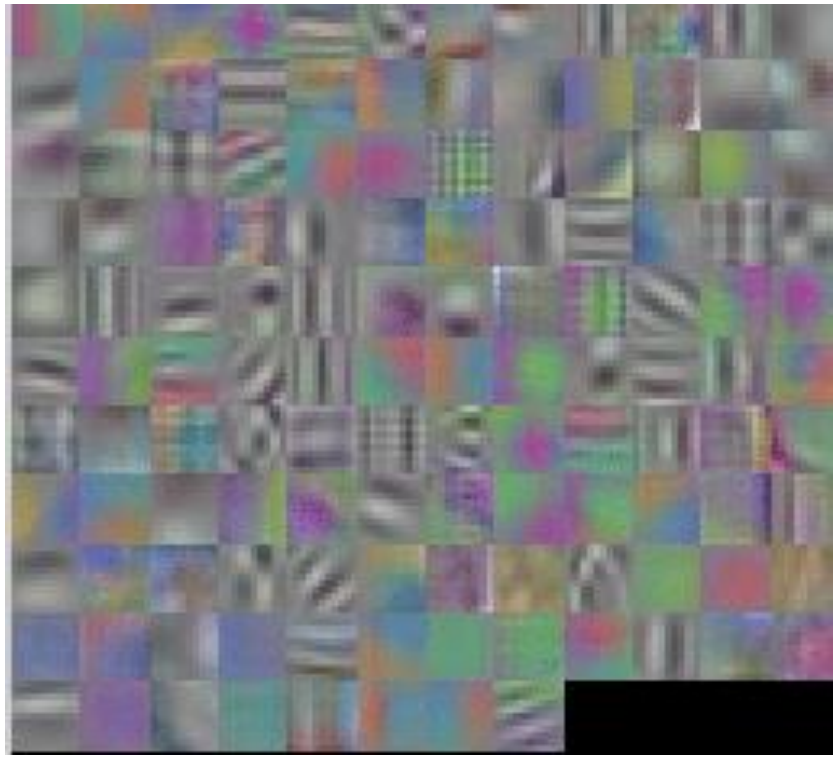
- Extend 2D Convs and pooling to 3D to slowly fuse temporal information throughout the model
- **Slide the kernels in both space and time**



Input: T x 3 x H x W



Weights: C x t x 3 x h x w



First layer filters
3(rgb) x 4 (t) x 5 (h) x 5 (w)

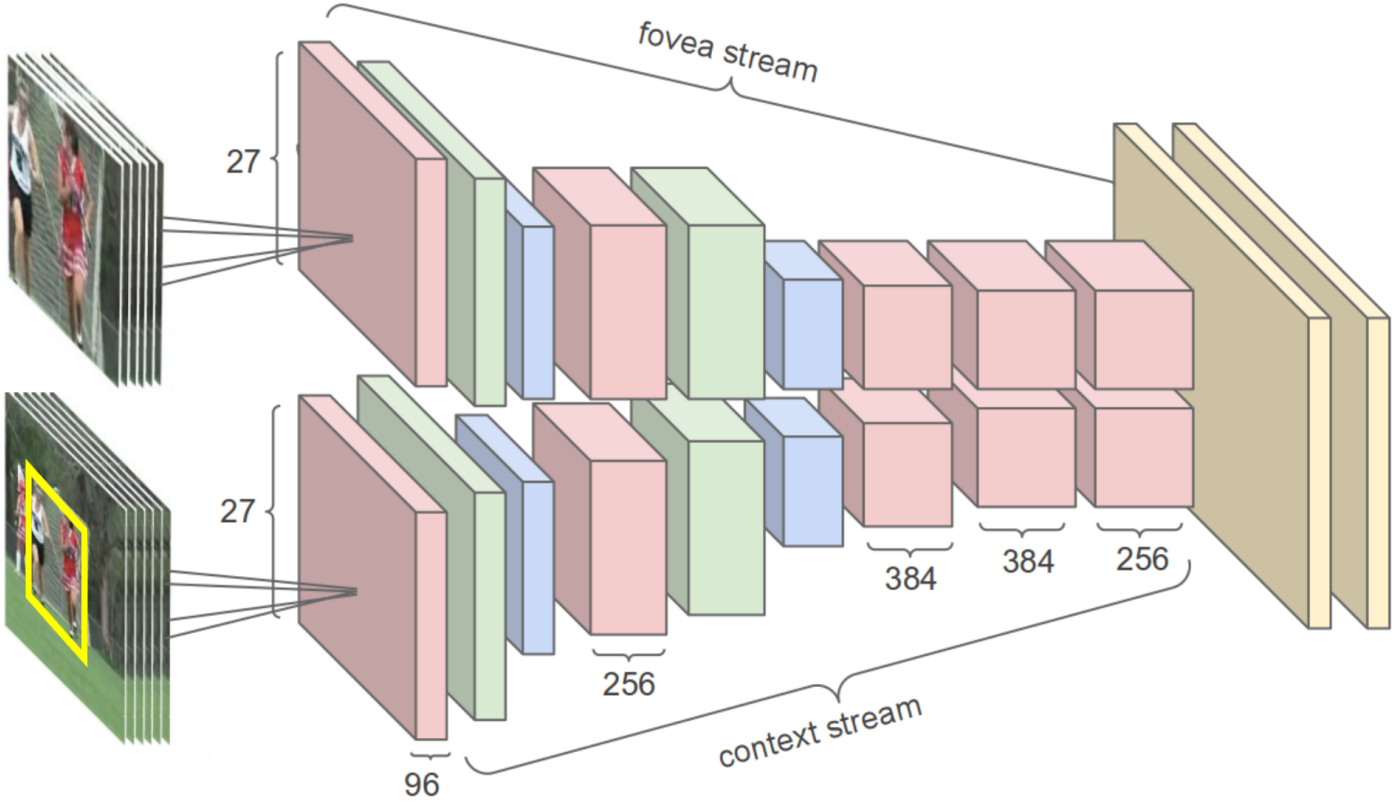
Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Models for Videos: Multi-scale

How can we reduce computational cost while maintaining accuracy?

Reduce video resolution → lower performance

Reduce network's capacity → lower performance



- Context stream ***(low res)***: process low res video frames (H/2, W/2)
- Fovea sterm ***(high res)***: process a (H/2, W/2) crop from the original resolution

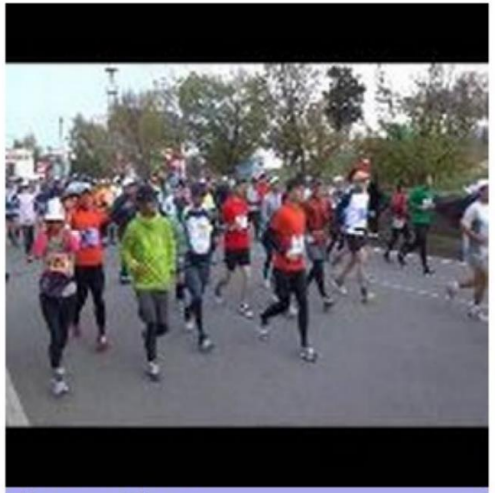
Reduce the input dimentionalty by half

Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Action classification -- Sports-1M



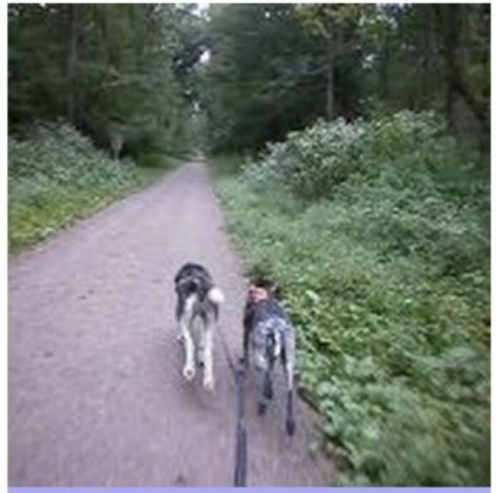
track cycling
cycling
track cycling
road bicycle racing
marathon
ultramarathon



ultramarathon
ultramarathon
half marathon
running
marathon
inline speed skating



heptathlon
heptathlon
decathlon
hurdles
pentathlon
sprint (running)



bikejoring
mushing
bikejoring
harness racing
skijoring
carting



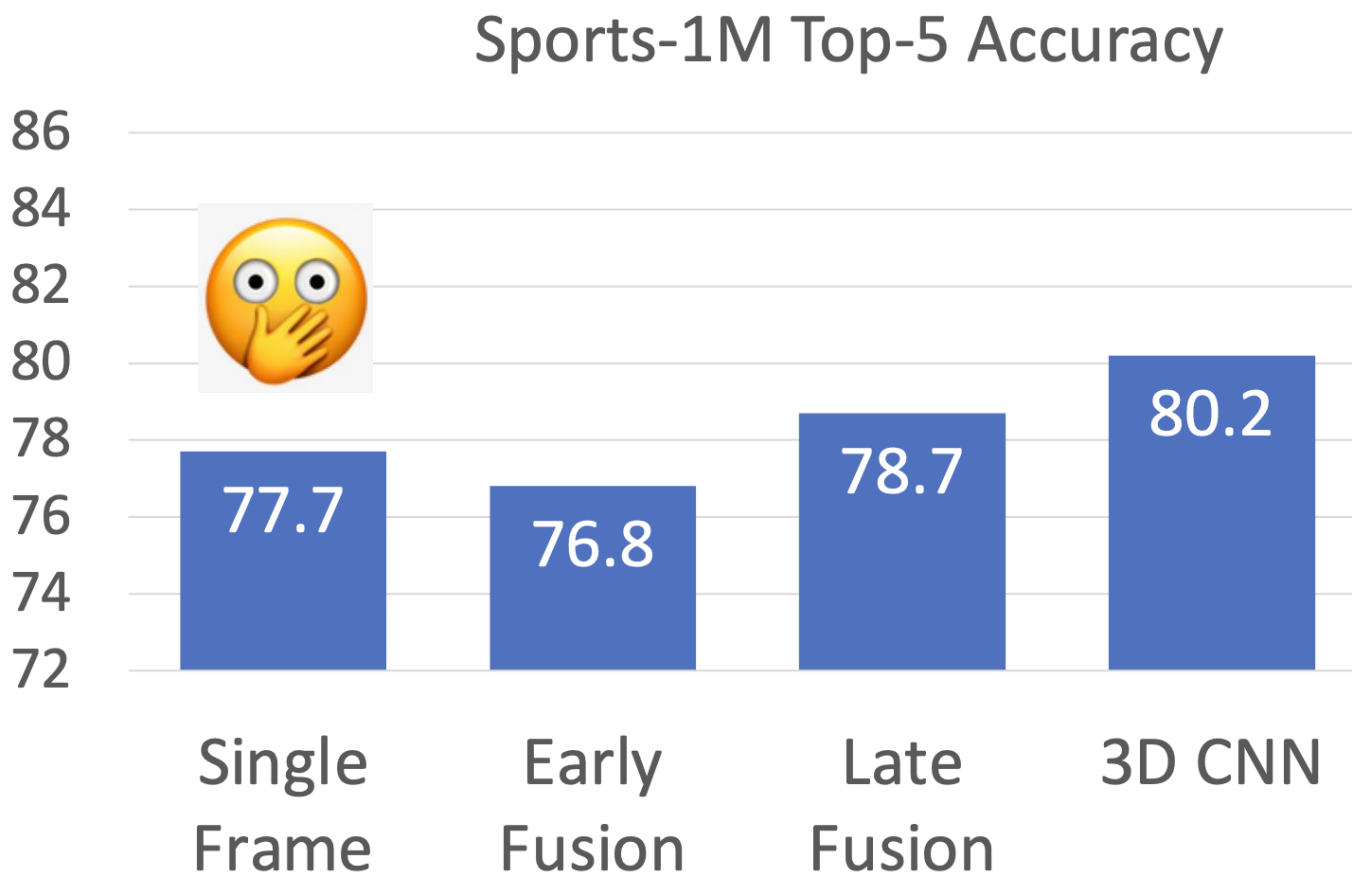
longboarding
longboarding
aggressive inline skating
freestyle scootering
freeboard (skateboard)
sandboarding

- **1 million YouTube videos**
- **Fine grained labels for 487 different types of sports**

- Ground truth
- Correct prediction
- Incorrect prediction

Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014

Action classification -- Sports-1M



Single frame: a shockly powerful baseline

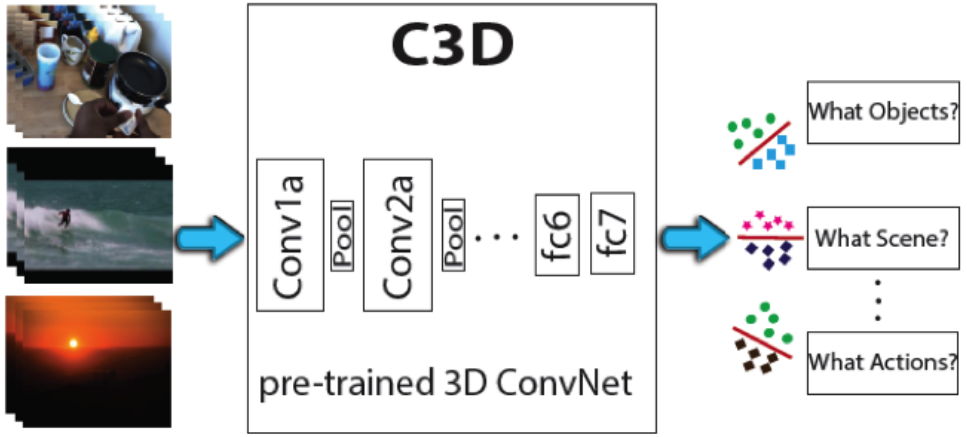
This is from 2014…

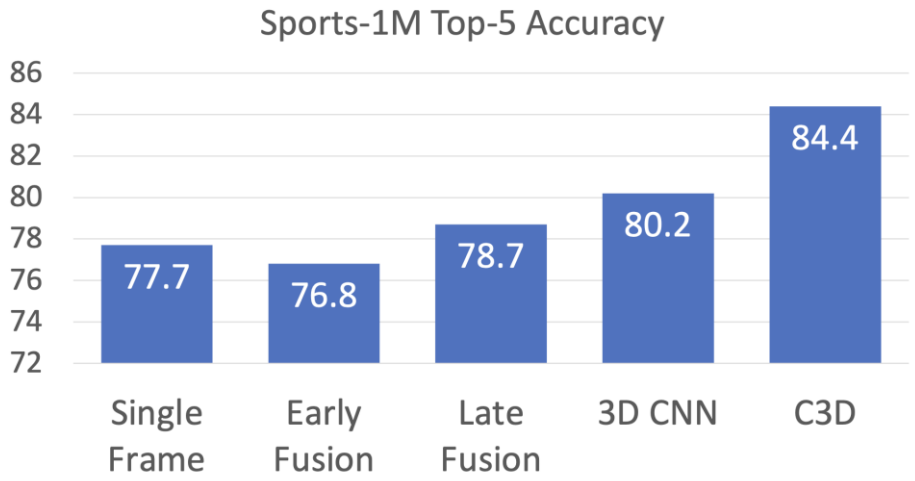
Karpathy et. al., Large-scale Video Classification with Convolutional Neural Networks, CVPR, 2014 Slide credit: Justin Johnson, EECS 498-007

Models for Videos: C3D (Convolutional 3D)

- 3D CNN that uses all 3x3x3 Convs and 2x2x2 poolings
- The “VGG” of 3D CNNs

- Transfer learning: extract learned video features, train a simple linear classifier for various tasks



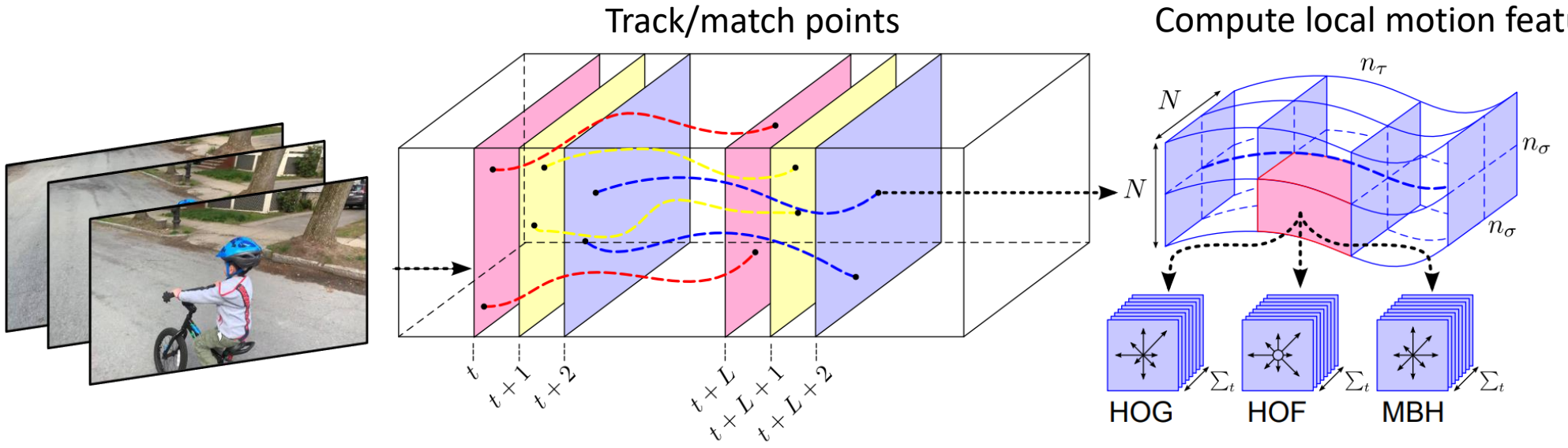


- Problem: 3D convs are VERY expensive!
  C3D on small inputs takes 3x VGG and 56x AlexNet FLOPs

Tran et al, “Learning Spatiotemporal Features with 3D Convolutional Networks”, ICCV 2015

Non-deep learning video classification

Motion is the most informative cue for action recognition → design hand crafted motion features:



Aggregate local motion features to compute a global representation of the video → linear SVM for action recognition

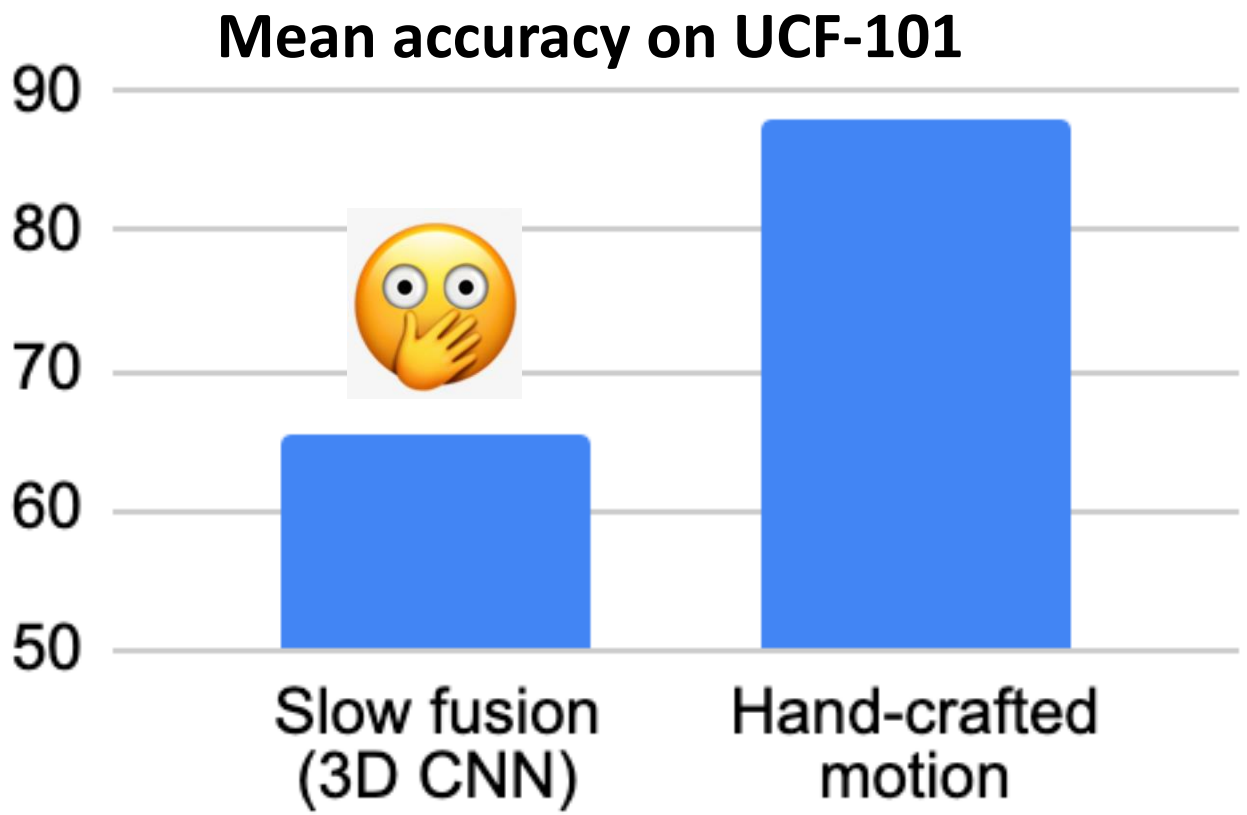
MODEL MOTION EXPLICITLY

Wang et. al., Dense trajectories and motion boundary descriptors for action recognition, 2013

Peng et. al., Bag of Visual Words and Fusion Methods for Action Recognition: Comprehensive Study and Good Practice, 2014

Non-deep learning video classification

Motion is the most informative cue for action recognition → hand crafted motion features:

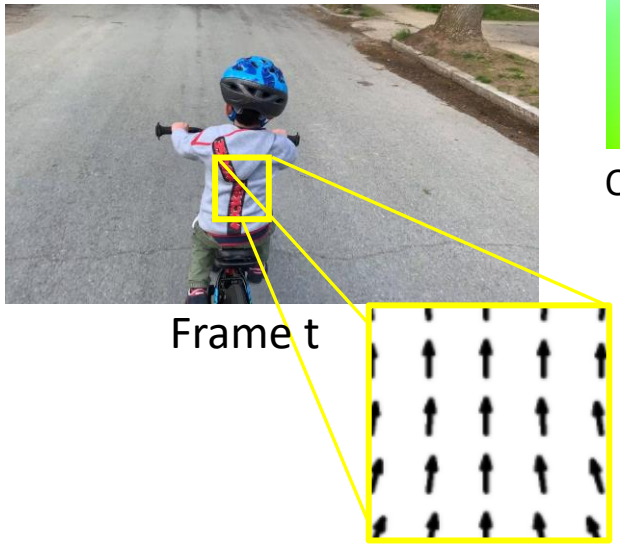


Explicitly modeling motion in deep-based models

Optical flow: For each pixel in frame t, determines its corresponding pixel in frame t+1



Frame t+1



Frame t

Optical flow provides **local motion cues**



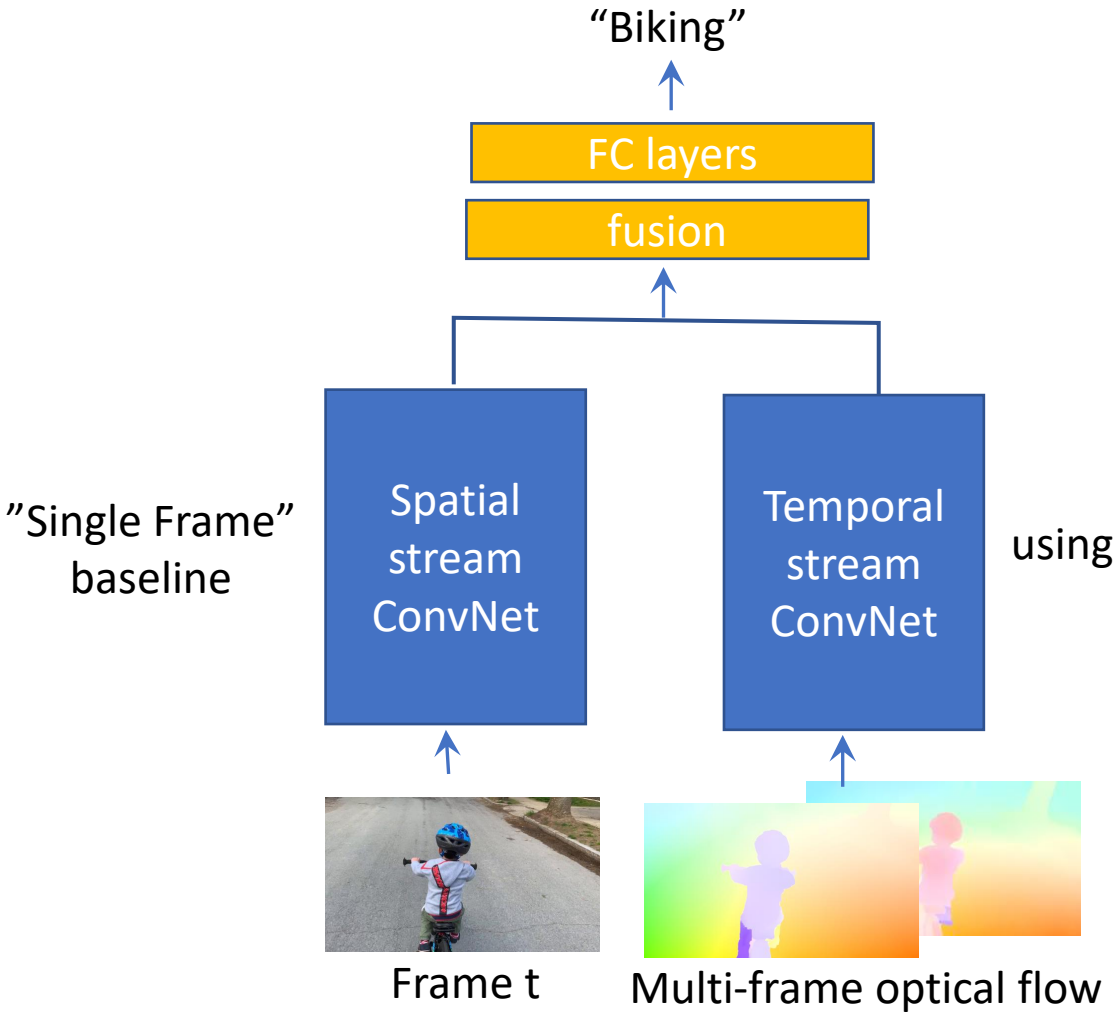
Optical flow between two frames



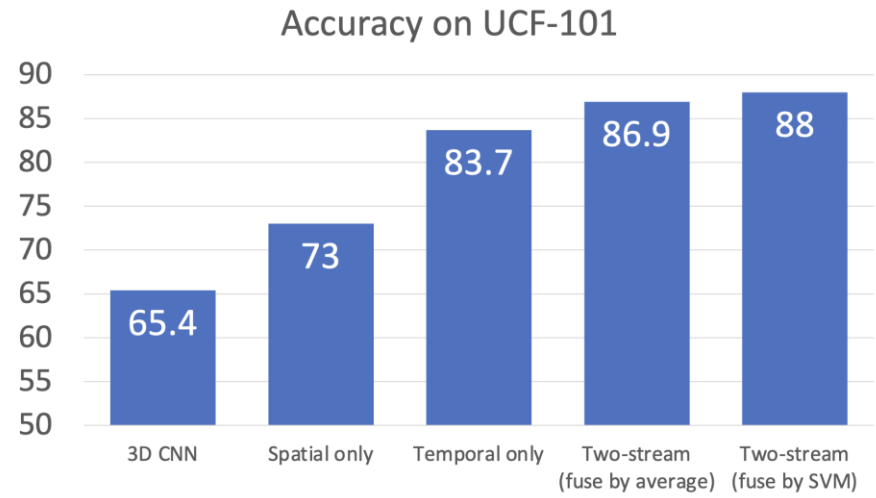
Color wheel
Saturation = mag.
Color = angle

Two Stream Networks: modeling motion explicitly

Idea: separate motion (multi-frame) from static appearance (single frame)



"Single Frame" baseline



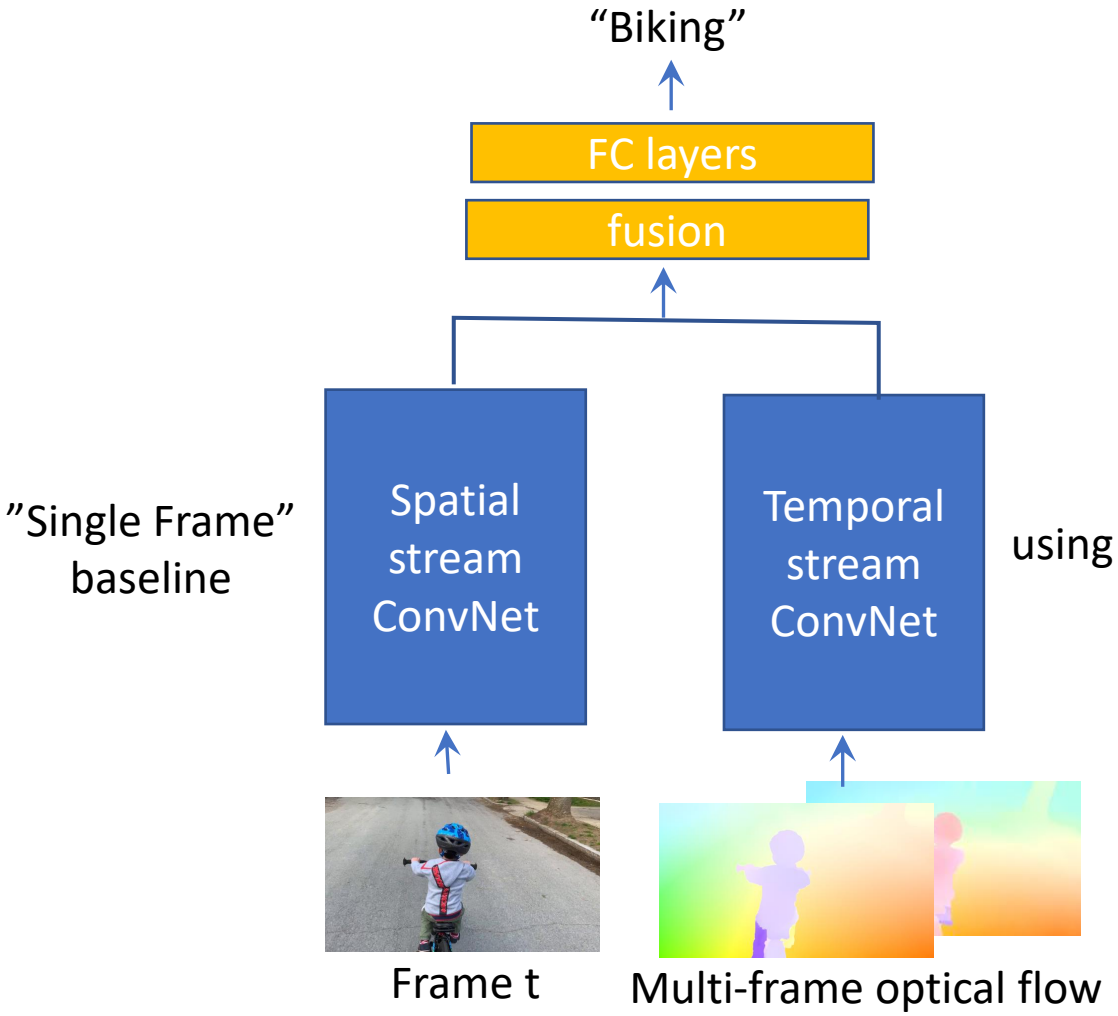
using "Early Fusion" baseline

Precomputed flow fields between concuective frames

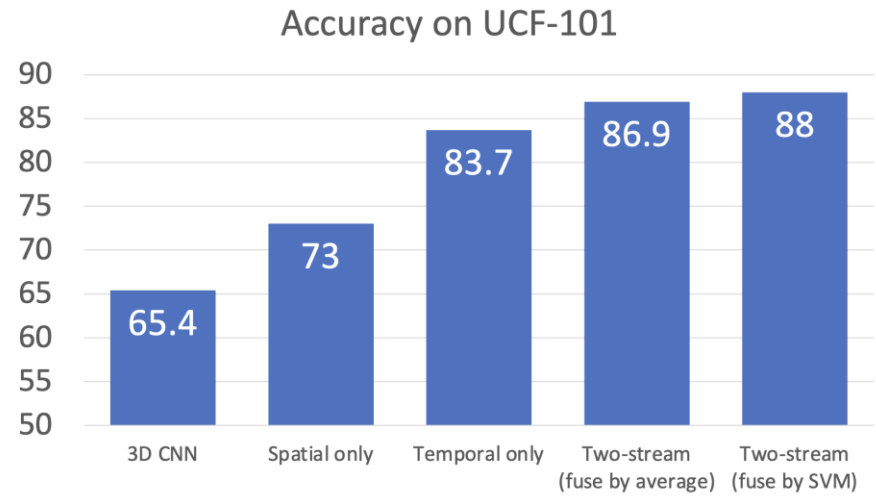
Simonyan and Zisserman, Two-Stream Convolutional Networks for Action Recognition in Videos, NIPS 2014

Two Stream Networks: modeling motion explicitly

Idea: separate motion (multi-frame) from static appearance (single frame)



"Single Frame" baseline



using "Early Fusion" baseline

Precomputed flow fields between concuective frames

Simonyan and Zisserman, Two-Stream Convolutional Networks for Action Recognition in Videos, NIPS 2014

Additional models

Inflating 2D networks to 3D (I3D)

Take an existing 2D CNN model → convert it to a 3D CNN model

Transfer the weights from 2D and 3D

Carreira and Zisserman, “Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset”, CVPR 2017

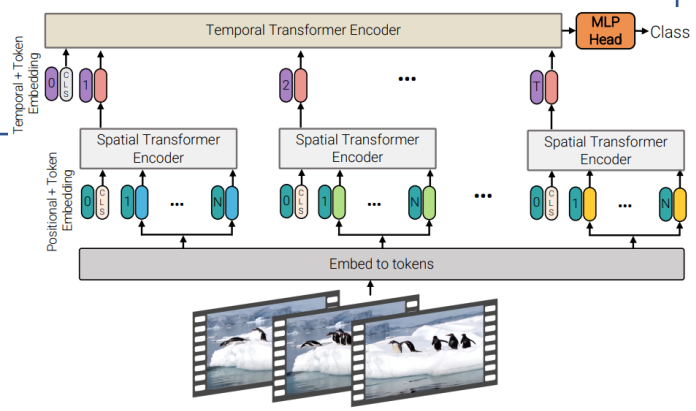
Long range temporal processing

Use LSTMs and RNNs to model long range temporal information

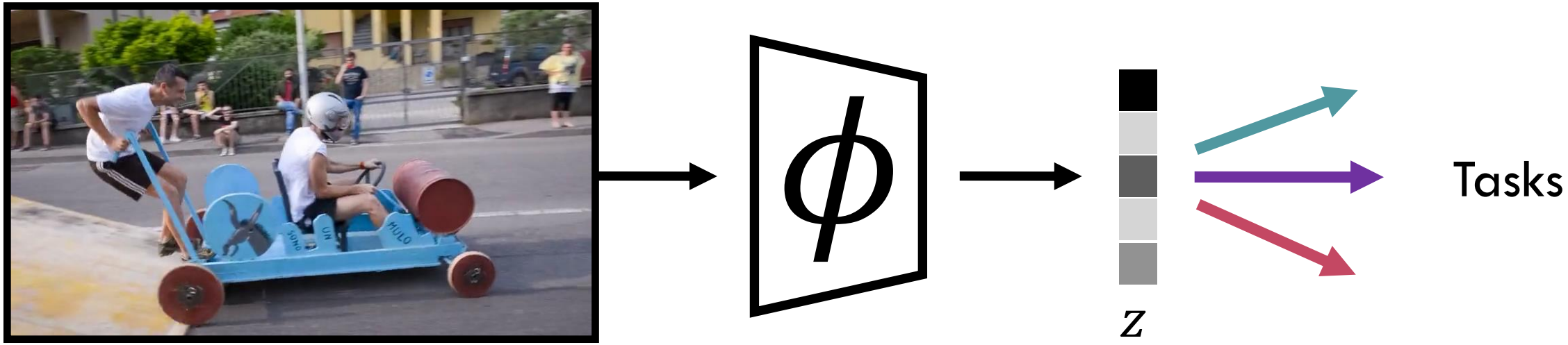
Baccouche et al, "Sequential Deep Learning for Human Action Recognition”, 2011 Donahue et al, “Long-term recurrent convolutional networks for visual recognition and description”, CVPR 2015

Long range temporal processing

Self attention, non-local networks, Transformers



Self-Supervision in Videos

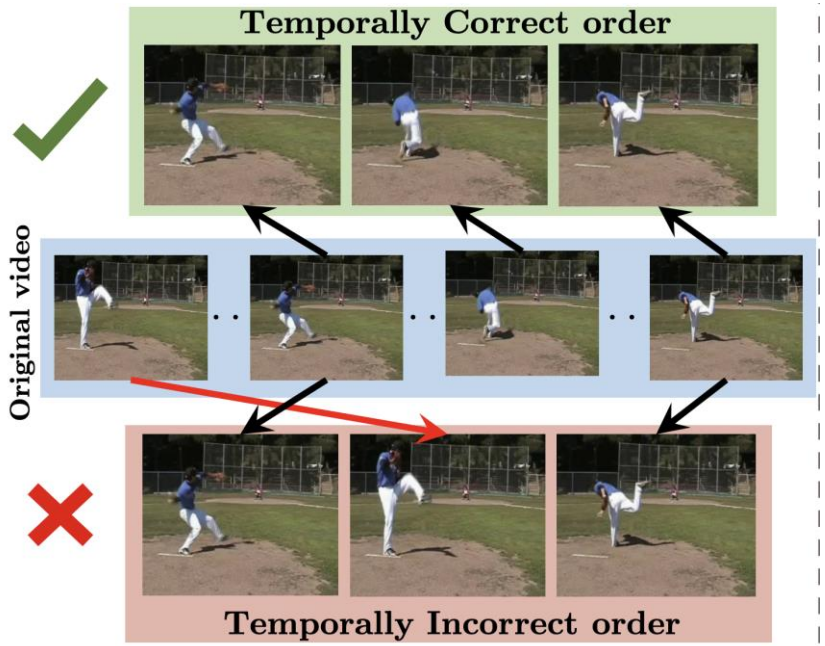


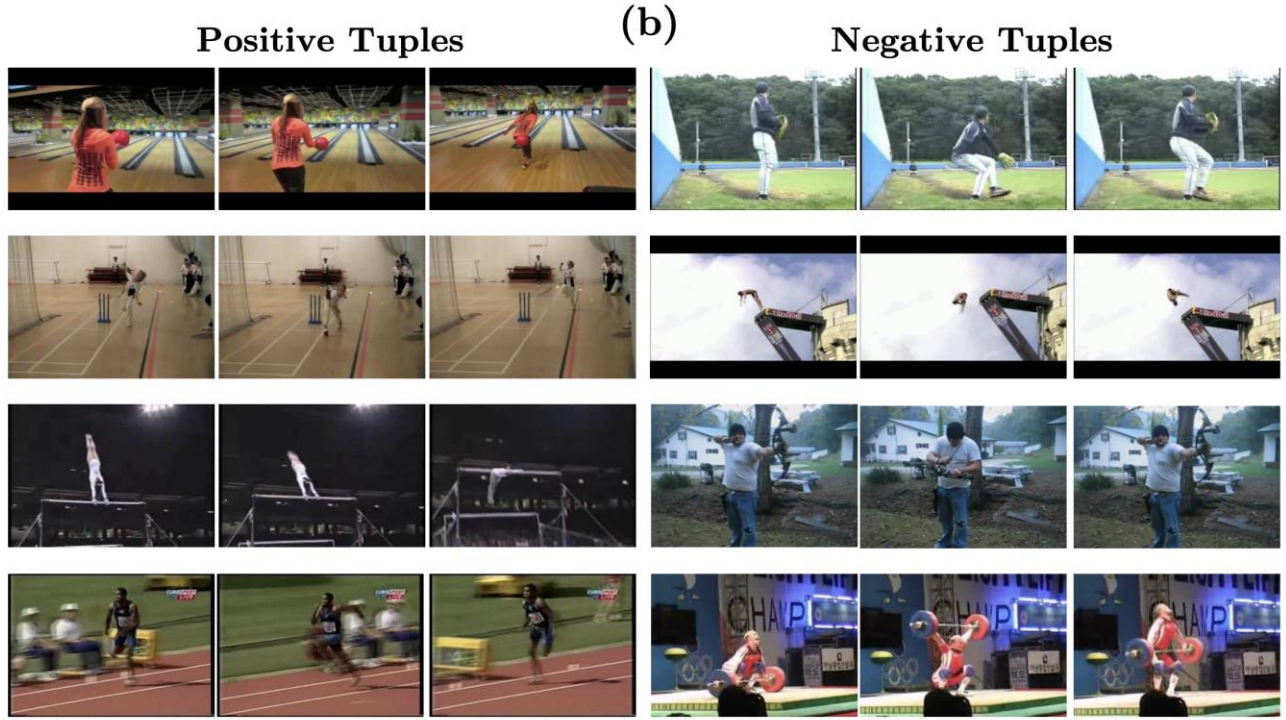
- Temporal order
- Cycle consistency
- Video Speedup
- Video colorization

Video: https://ajabri.github.io/timecycle/

Self-Supervision in Videos: frame ordering

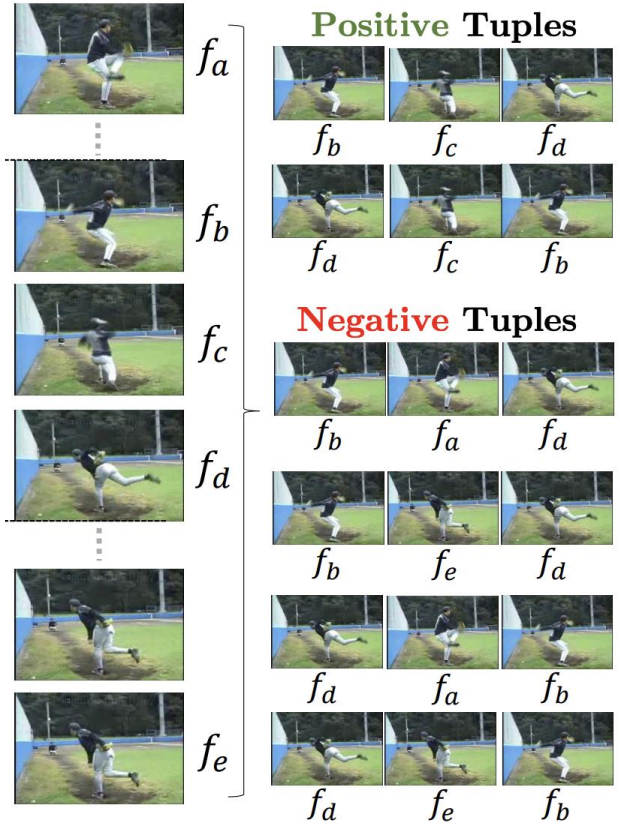
Training data: shuffled video frames, original video frames
Pretext task: predict if the frames are in the correct temporal order (binary classification task)



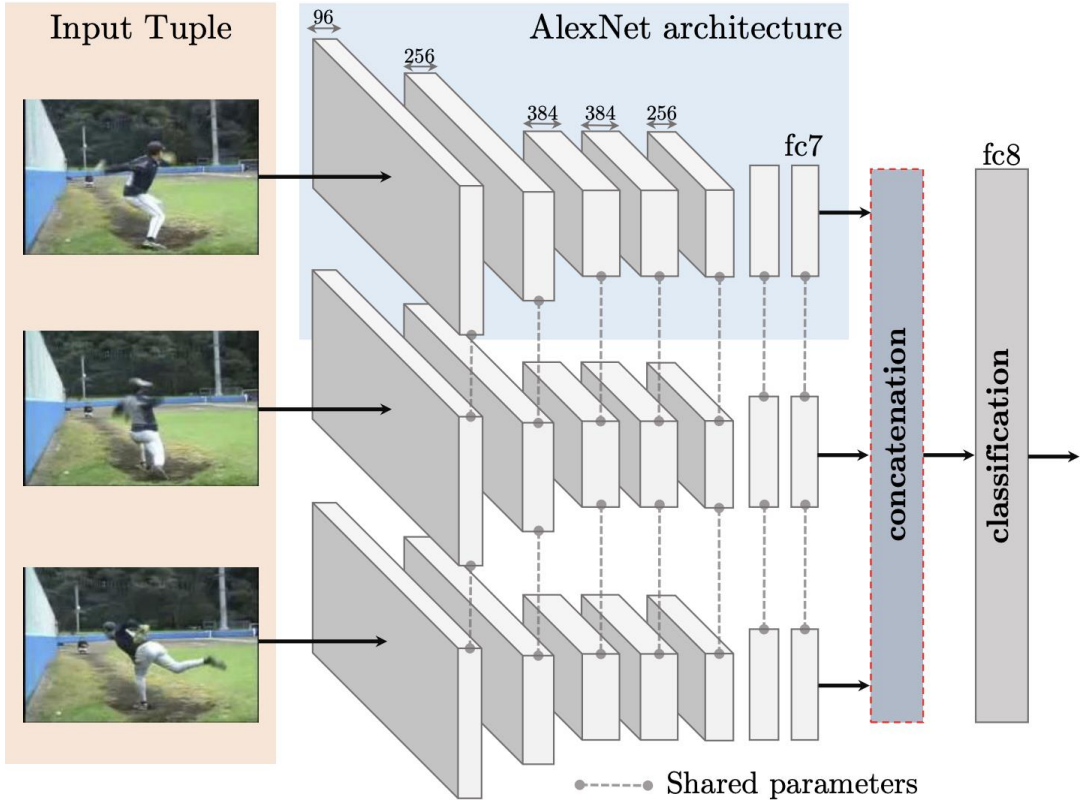


Misra et. al., Shuffle and Learn: Unsupervised Learning using Temporal Order Verification, ECCV 2016

Self-Supervision in Videos: frame ordering



Generating positive and negative examples

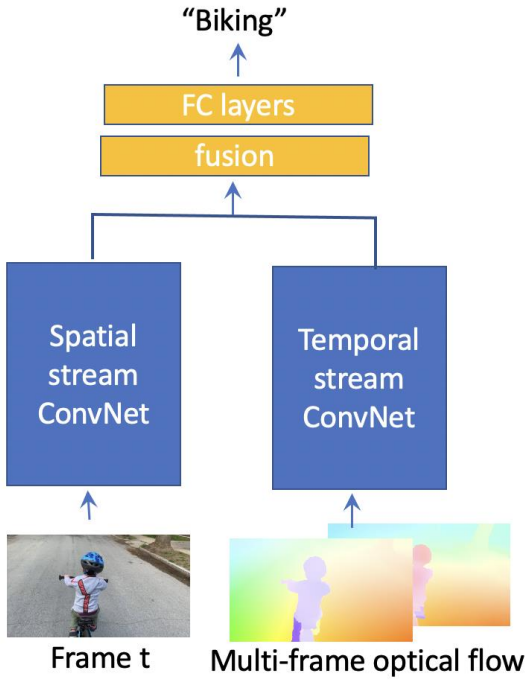


Triplet Siamese network for sequence verification

Misra et. al., Shuffle and Learn: Unsupervised Learning using Temporal Order Verification, ECCV 2016

Self-Supervision in Videos: frame ordering

Transfer learning: fine-tune spatial stream for video classification



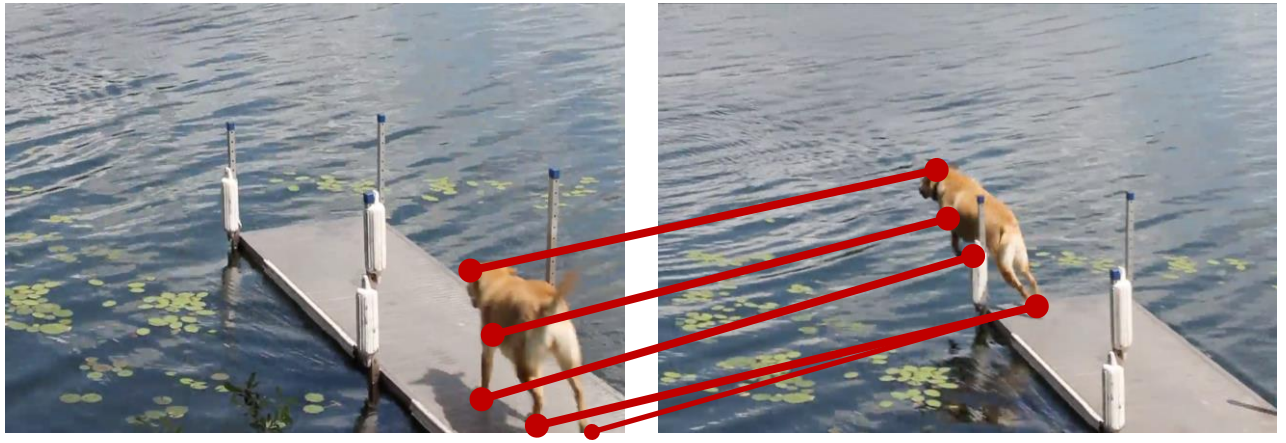
Dataset	Initialization	Mean Accuracy
UCF101	Random	38.6
	(Ours) Tuple verification	**50.2**
HMDB51	Random	13.3
	UCF Supervised	15.2
	(Ours) Tuple verification	**18.1**

Misra et. al., Shuffle and Learn: Unsupervised Learning using Temporal Order Verification, ECCV 2016

Self-Supervision in Videos: Learning correspondence

Ultimate goal: Correspondence





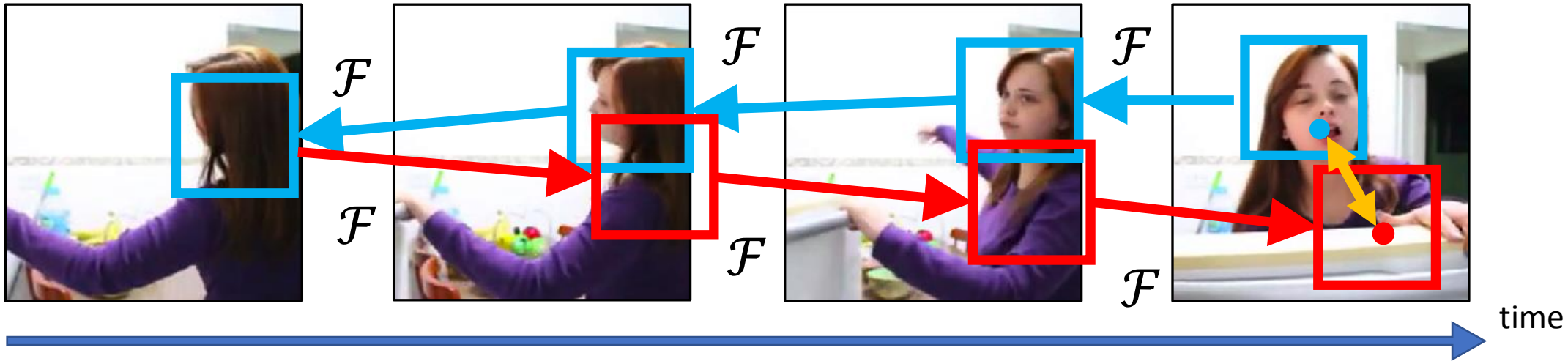
Wang and Efros, Learning Correspondence from the Cycle-consistency of Time, CVPR 2019

Self-Supervision in Videos: Learning correspondence

Ultimate goal: Correspondence, without using off-the-shelf tracking methods

How to obtain supervision?

Supervision: Cycle-Consistency in Time



Track backwards in time

Track forwards, back to the future

Wang and Efros, Learning Correspondence from the Cycle-consistency of Time, CVPR 2019

Self-Supervision in Videos: Learning correspondence

Supervision: Cycle-Consistency in Time
Challenge: Occlusions



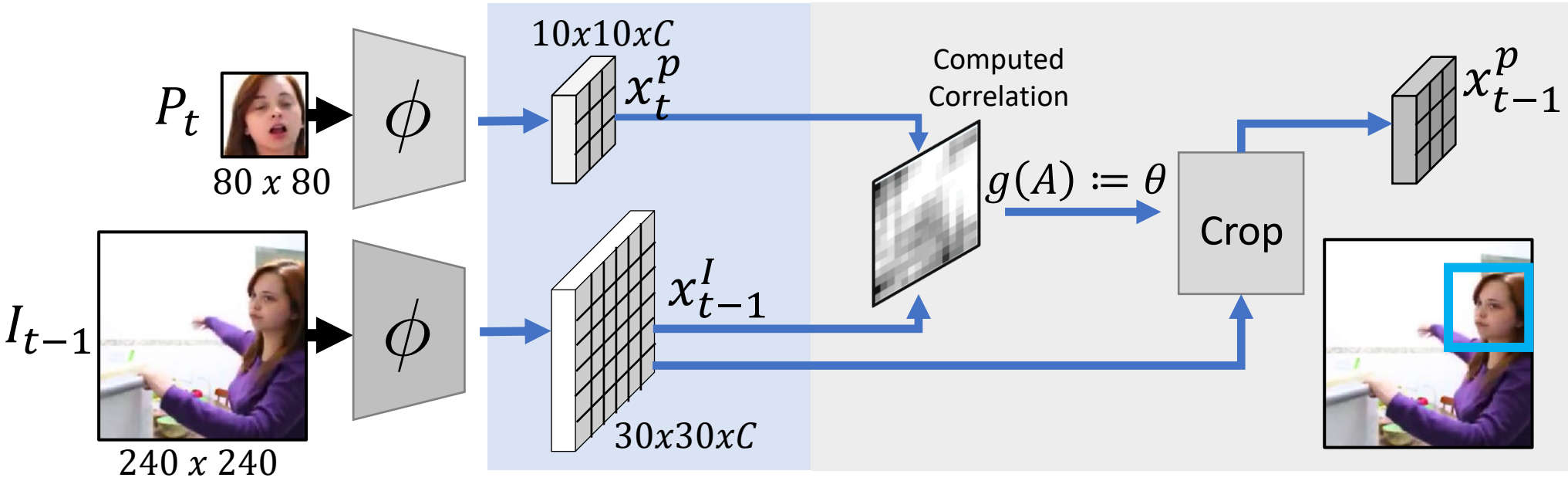
Skip-cycles: skipping occlusions

Wang and Efros, Learning Correspondence from the Cycle-consistency of Time, CVPR 2019

Self-Supervision in Videos: Learning correspondence

Differentiable tracker: densely match features in learned feature space

$$A(j,i) = \frac{\exp\left(x^I(j)^\intercal x^p(i)\right)}{\sum_j \exp\left(x^I(j)^\intercal x^p(i)\right)} \qquad A \in R^{900x100}$$

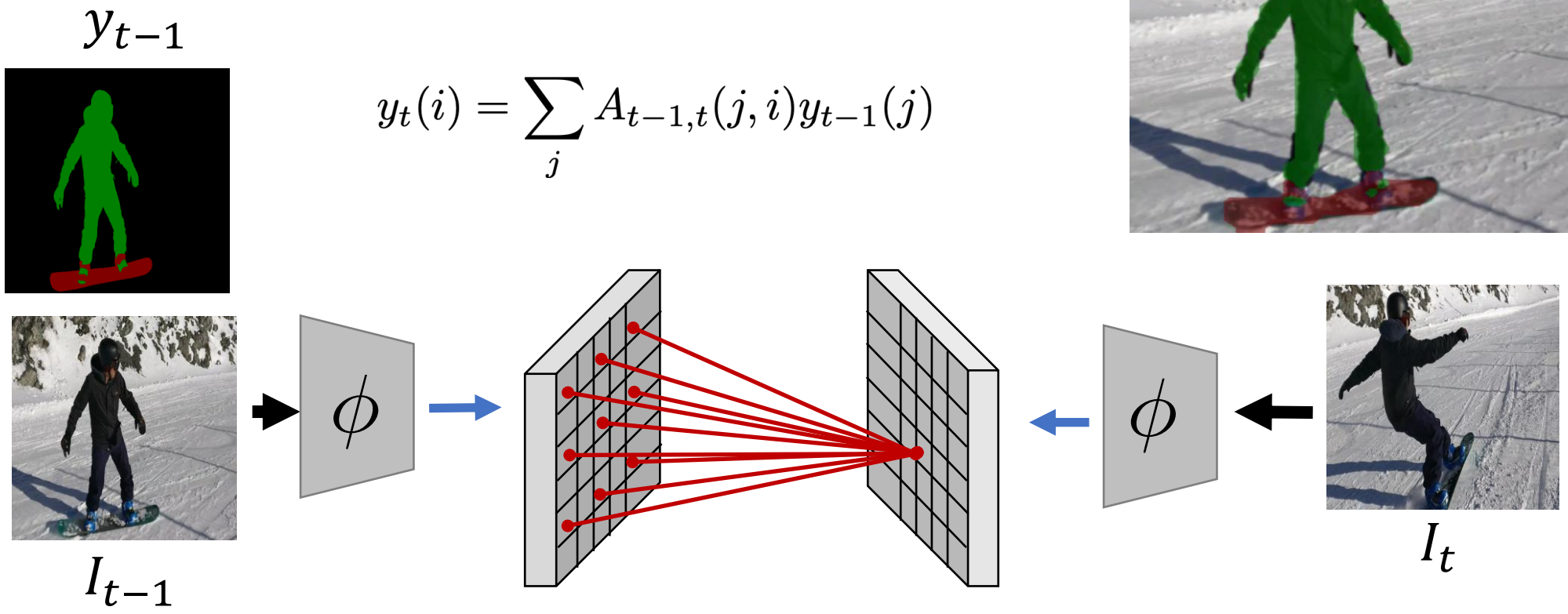


Wang and Efros, Learning Correspondence from the Cycle-consistency of Time, CVPR 2019

Self-Supervision in Videos: Learning correspondence

Test time: compute features to each frame, compute features affinity, propagate information using the affinities





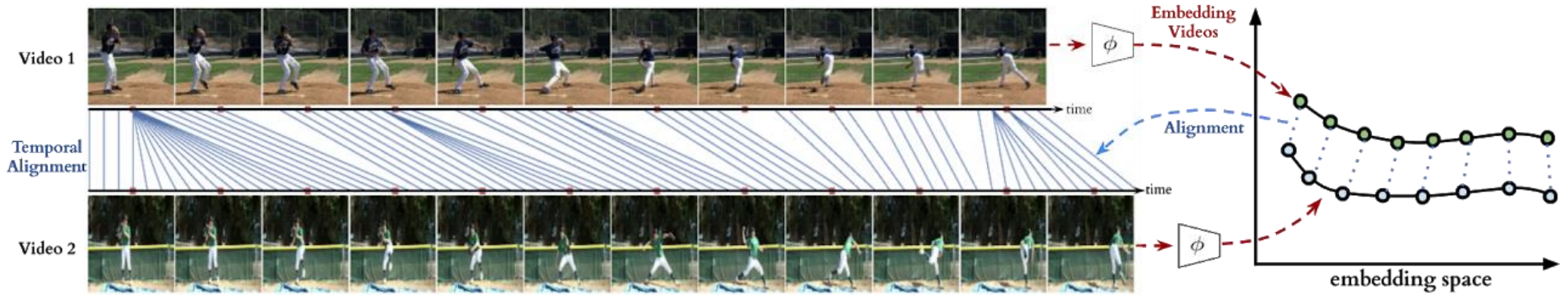
Wang and Efros, Learning Correspondence from the Cycle-consistency of Time, CVPR 2019

Self-Supervision in Videos: Learning correspondence

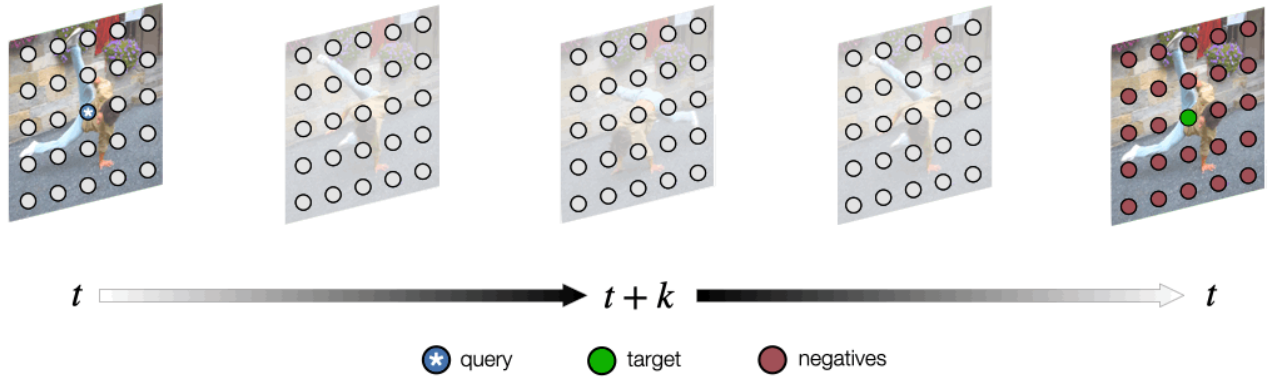


Wang and Efros, Learning Correspondence from the Cycle-consistency of Time, CVPR 2019

Self-Supervision in Videos: Temporal cycle consistency



Dwibed et. al. Temporal Cycle-Consistency Learning, CVPR'19



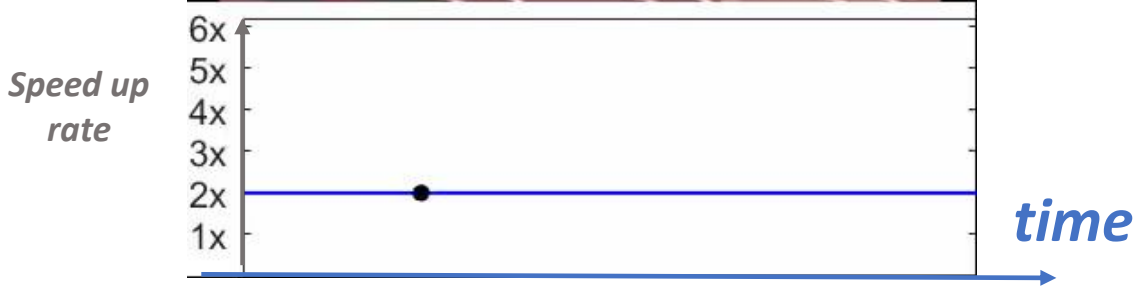
Jabri et. al, Space time correspondence as Contrastive Random Walk, NeurIPS 2020

Self-Supervision in Videos: Learning the Speediness in Videos

Ultimate goal: Watch video content faster by adaptively speeding up the video

Uniform Speed Up (2x)





Jittery, unnatural motions

Adaptive Speed Up (2x)





Same duration, more natural

Joint work with: Sagie Benaim, Ariel Ephrat, Oran Lang, Inbar Mosseri, Bill Freeman, Miki Rubinstein and Michal Irani, CVPR 2020

“Speediness” in Videos

Slower



Normal speed



Faster



Joint work with: Sagie Benaim, Ariel Ephrat, Oran Lang, Inbar Mosseri, Bill Freeman, Miki Rubinstein and Michal Irani, CVPR 2020

Self-Supervision in Videos: Learning the Speediness in Videos

Pretext task: Predict if a given video segment is sped up or not
Training data: sped up video segments + original video segments

Self supervised
training on Kinetics



Input segment
(30 frames)



Normal speed
or
Sped Up

"Learning and Using the Arrow of Time", Wei at. al, CVPR 2018



Self-Supervision in Videos: Learning the Speediness in Videos

Pretext task: Predict if a given video segment is sped up or not
Training data: sped up video segments + original video segments

Self supervised training on Kinetics



Input segment (30 frames)



Normal speed
or
Sped Up

Networks are lazy!



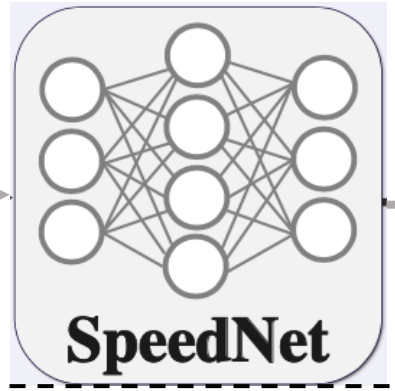
Learning properties of natural motion, avoid "easy cheats" → **very challenging!**

Self-Supervision in Videos: Learning the Speediness in Videos

Pretext task: Predict if a given video segment is sped up or not
Training data: sped up video segments + original video segments

Self supervised training on Kinetics



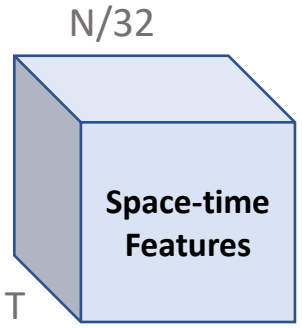


Normal speed
or
Sped Up



3D Conv Network

** Based on S3D-G*



Pooling

(spatial max, temporal average)



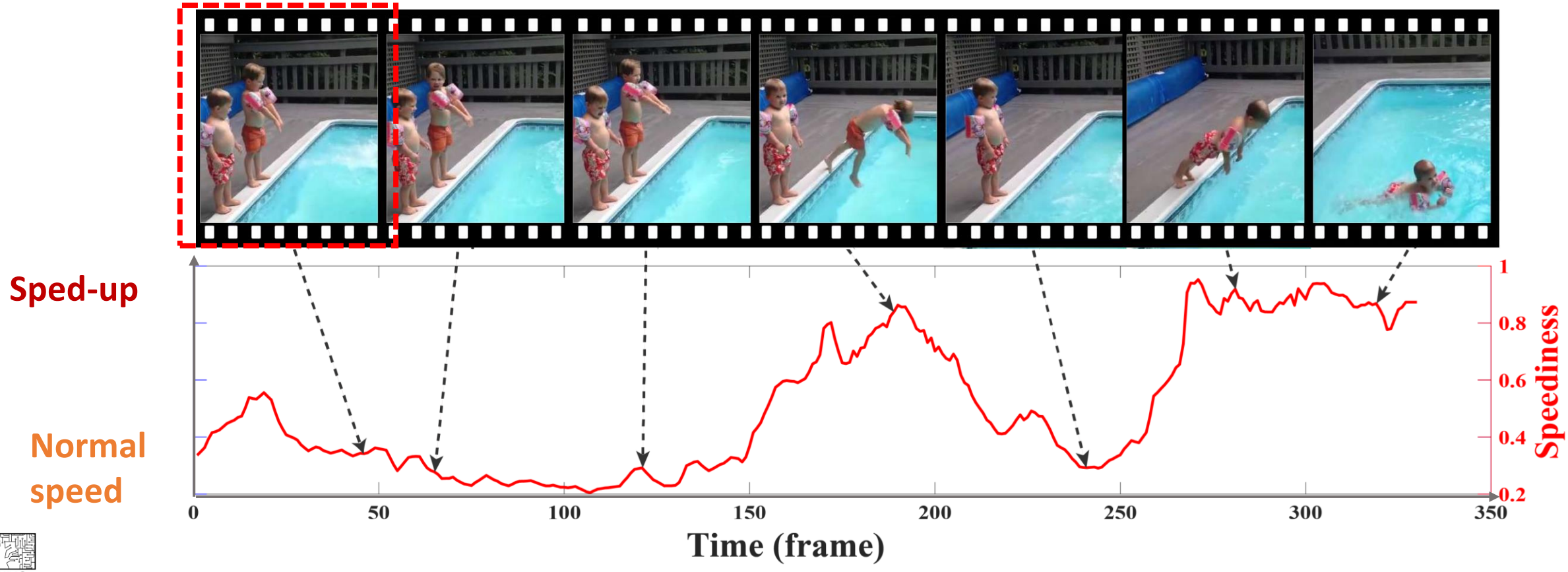
1x1 Conv

Normal Speed
or
Sped Up

** "Rethinking spatiotemporal feature learning: Speed-accuracy trade-offs in video classification",* Saining Xie, Chen Sun, Jonathan Huang, Zhuowen Tu, and Kevin Murphy, ECCV'18.

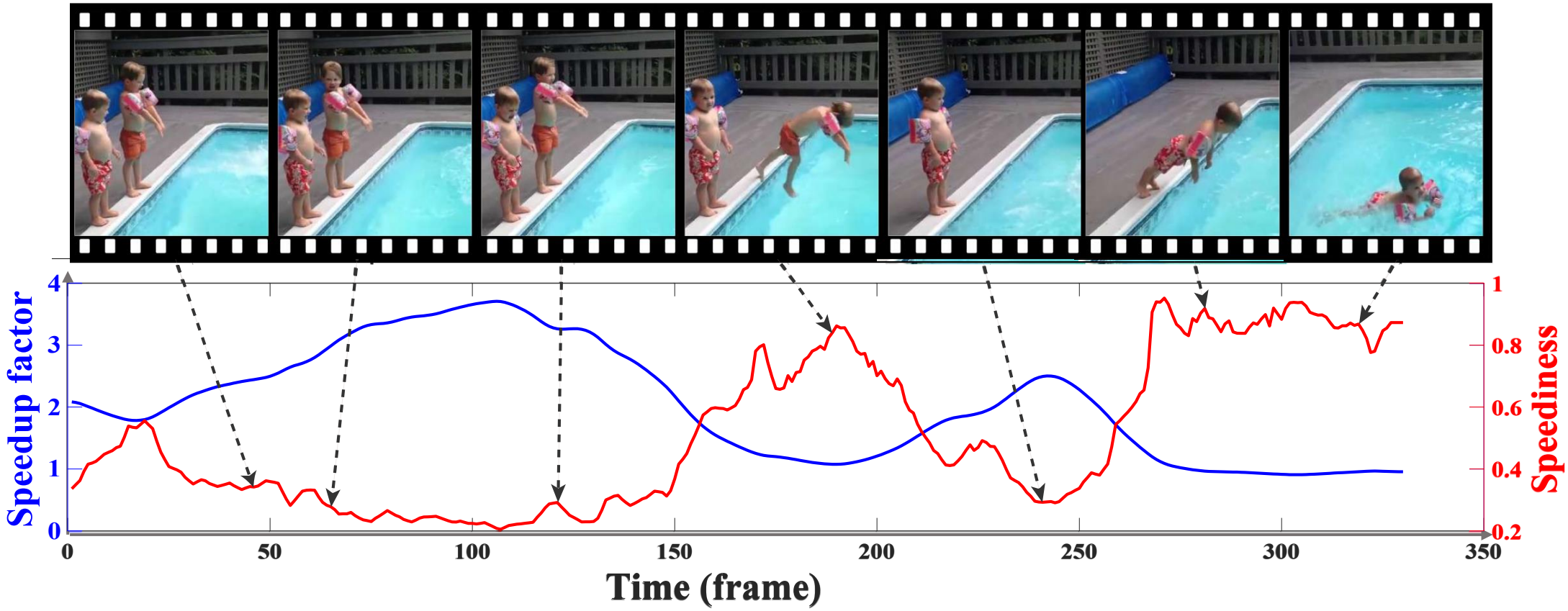
Self-Supervision in Videos: Learning the Speediness in Videos

Inference: sliding window → prediction for every frame

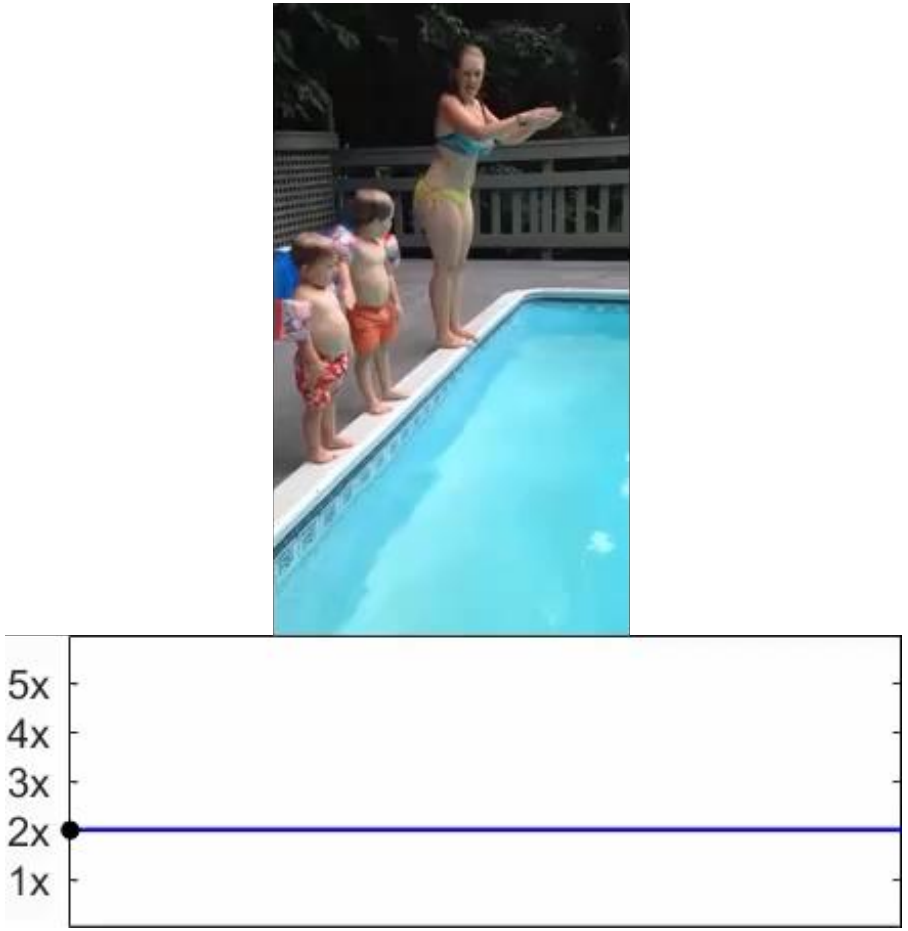


Self-Supervision in Videos: Learning the Speediness in Videos

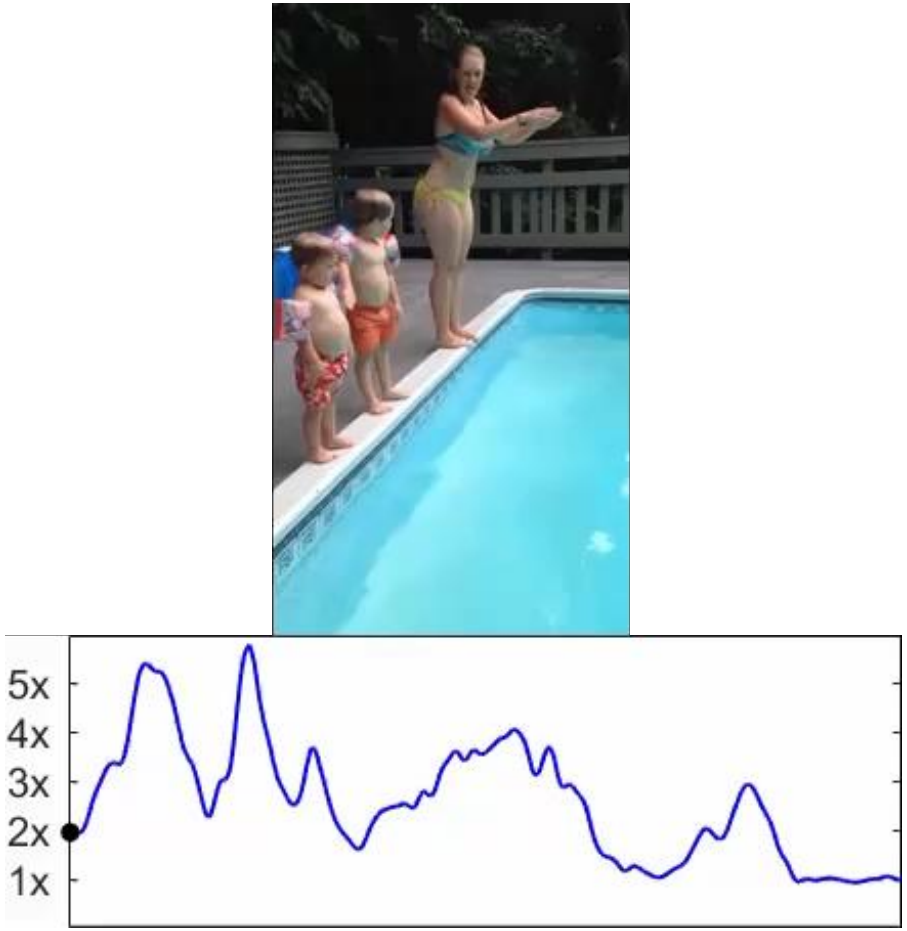
From "Speediness" to Speedup factor:
Low speediness → speedup more
High speediness→ speedup less



Learning the Speediness in Videos: Adaptive Video Speedup



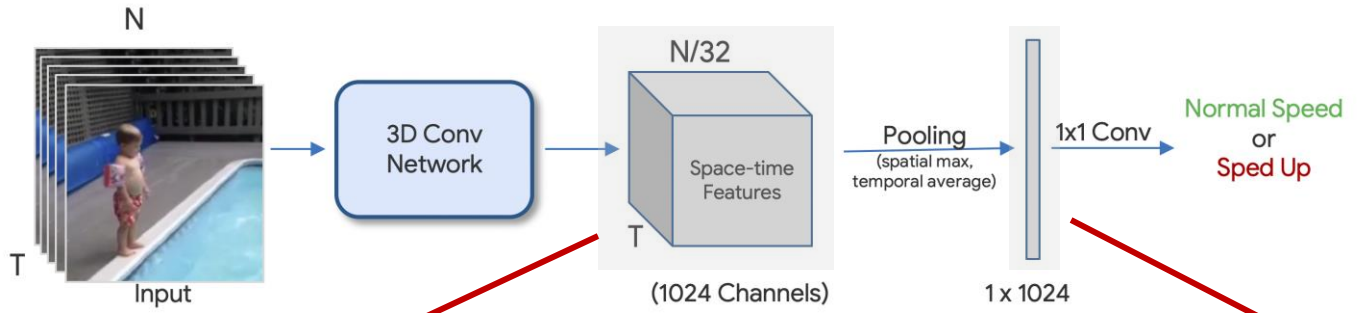
Uniform Speedup 2x



Adaptive Speedup 2x (ours)

Learning the Speediness in Videos: Transfer Learning

Pre-trained SpeedNet

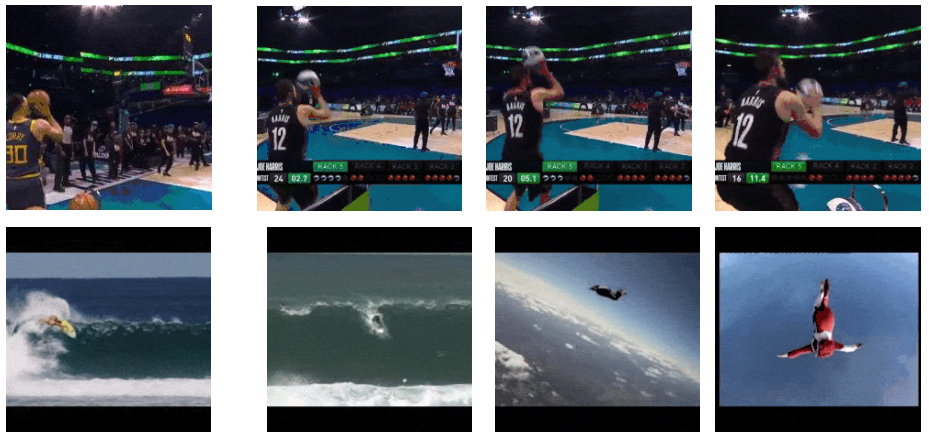


Self Supervised Action Recognition

Method	Architecture	Initialization: Supervised accuracy UCF101	HMDB51
Random init	S3D-G	73.8	46.4
ImageNet inflated	S3D-G	86.6	57.7
Kinetics supervised	S3D-G	96.8	74.5
CubicPuzzle [19]	3D-ResNet18	65.8	33.7
Order [40]	R(2+1)D	72.4	30.9
DPC [13]	3D-ResNet34	75.7	35.7
AoT [38]	T-CAM	79.4	-
SpeedNet (Ours)	S3D-G	**81.1**	**48.8**

Video Retrieval

Query | Retrieved top-3 results



Learning the Speediness in Videos: CAM visualizations



"Memory Eleven"
artistic video by Bill Newsinge



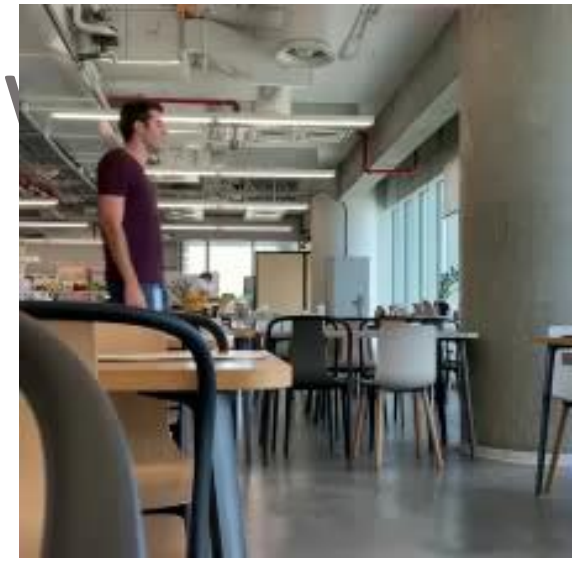
Our space-time
speediness visualization

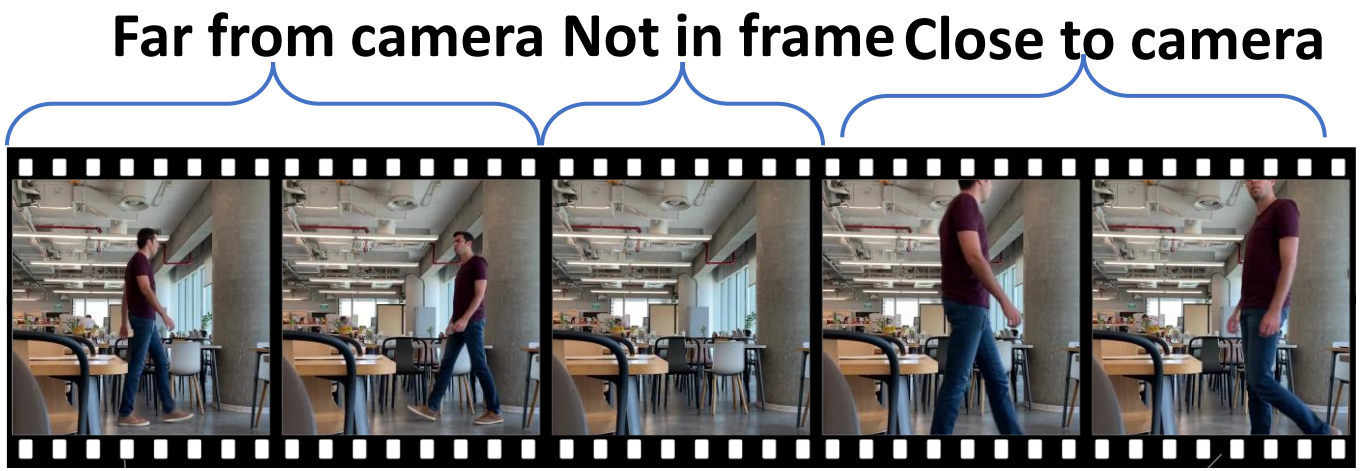
blue/green =
normal speed

yellow/orange =
slowed down

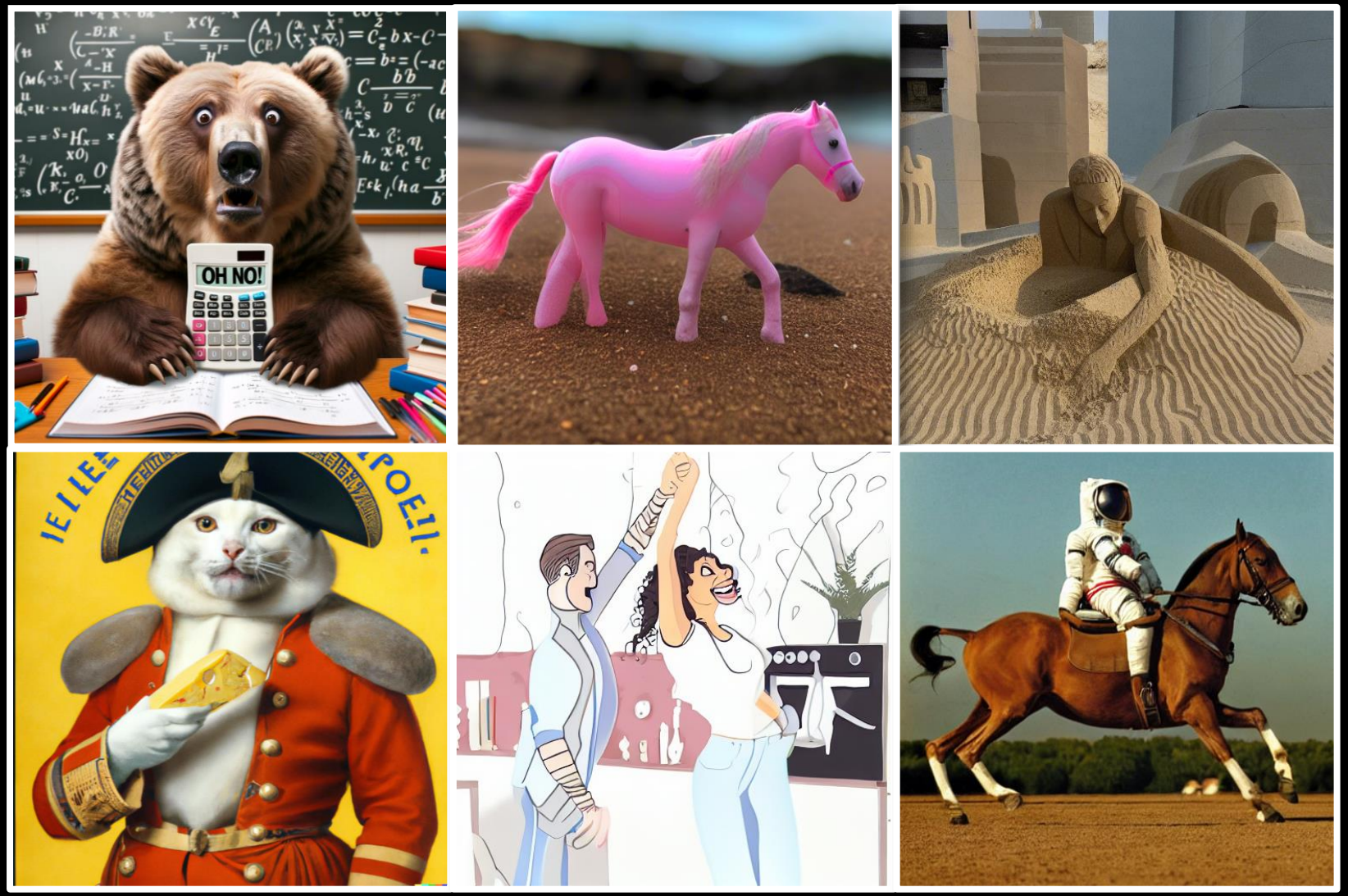
https://www.youtube.com/watch?v=djylS0Wi_Io

Speediness ≠ Magnitude of Optical Flo





The Generative AI Revolution



The Generative AI Revolution







Text-to-Image Model







What about videos?

A cat next to a dog



et al. Imagen Video. 2022


LUMIERE

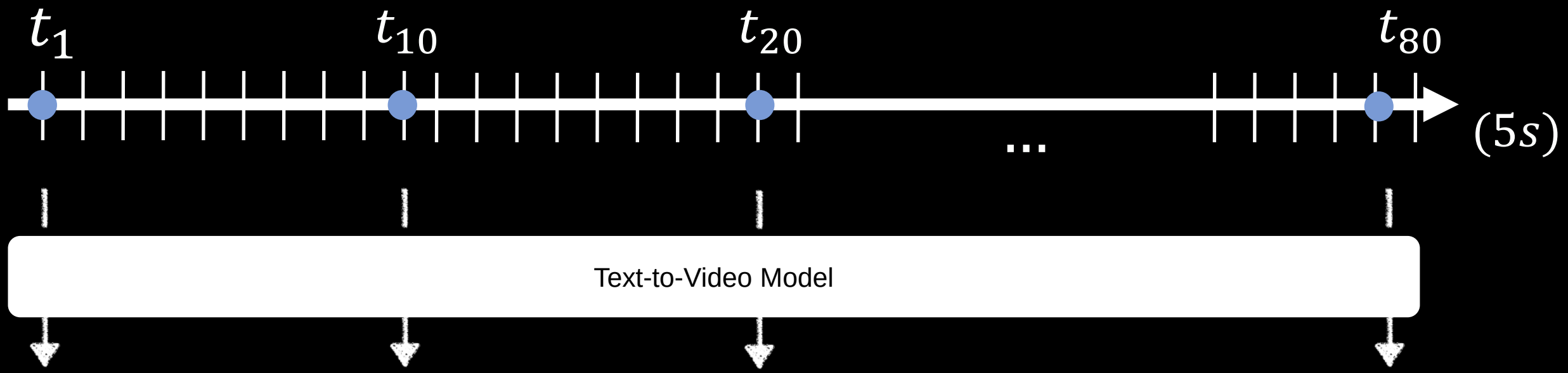
Processing video is expensive!

Common approach:

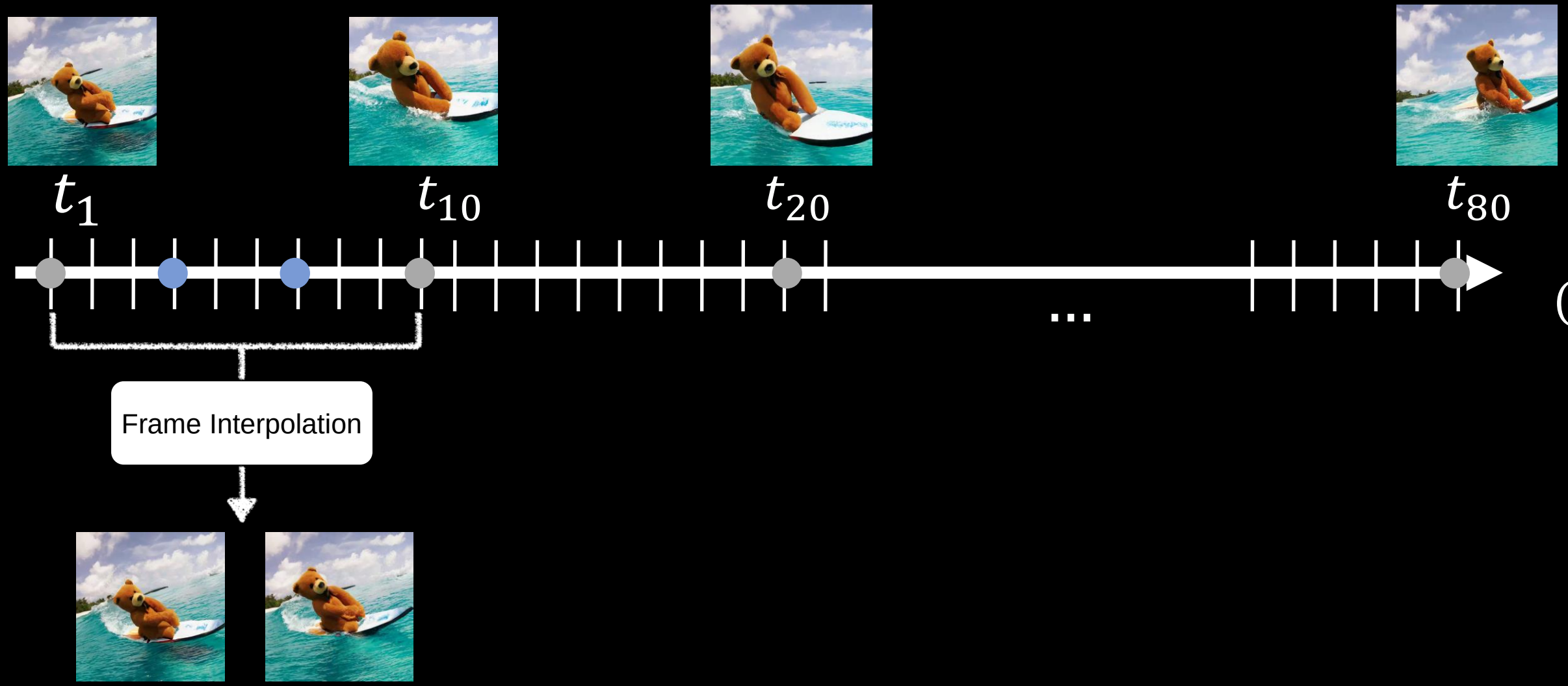
1) generate distant keyframes

2) fill-in missing content

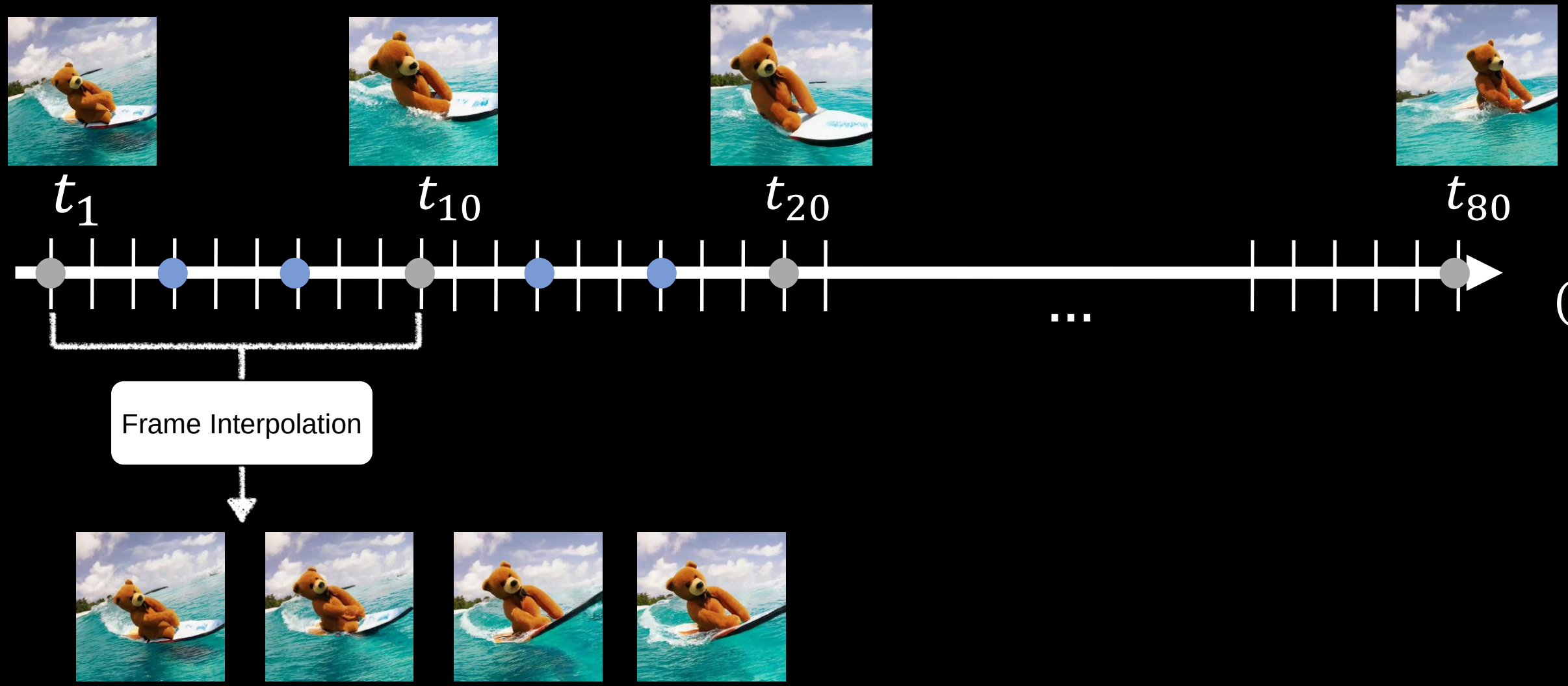
1) Generate distant keyframes



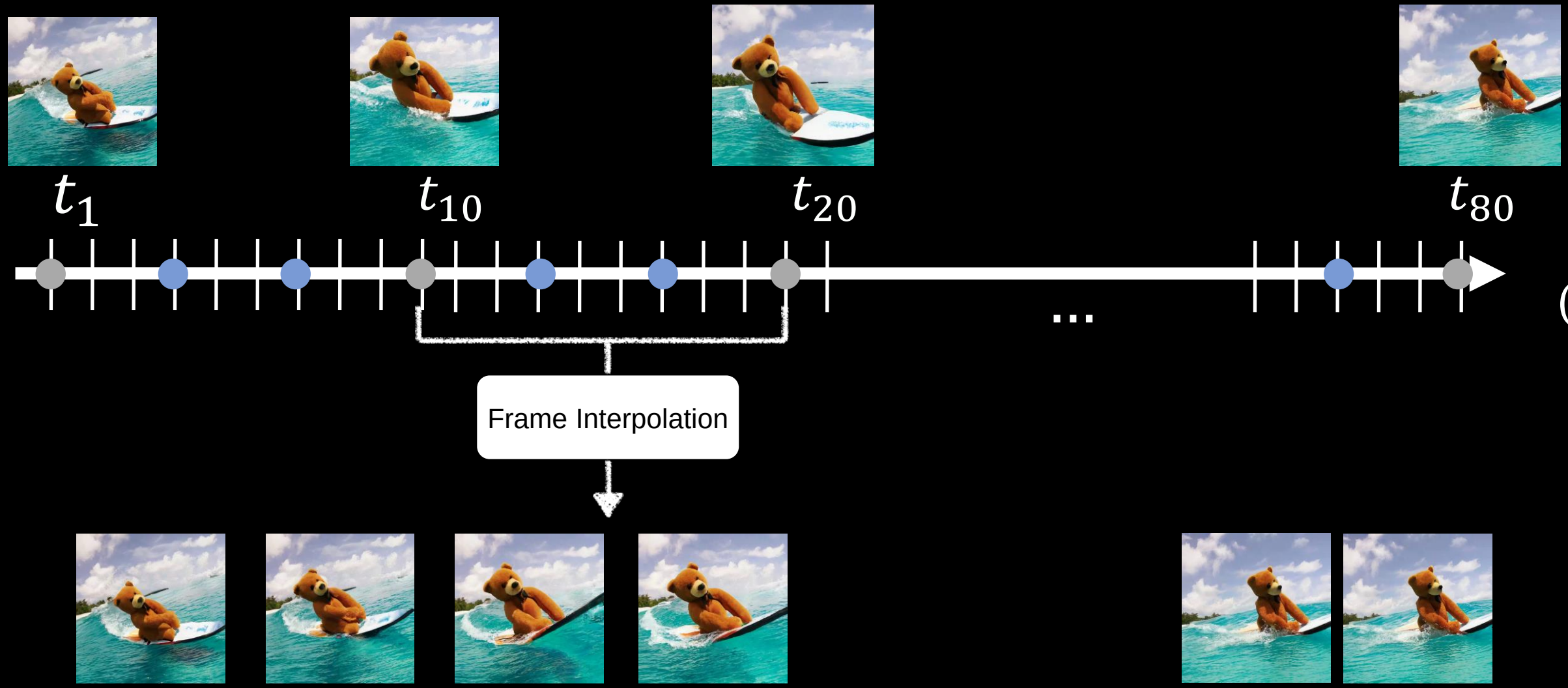
2) Fill-in missing frames



2) Fill-in missing frames

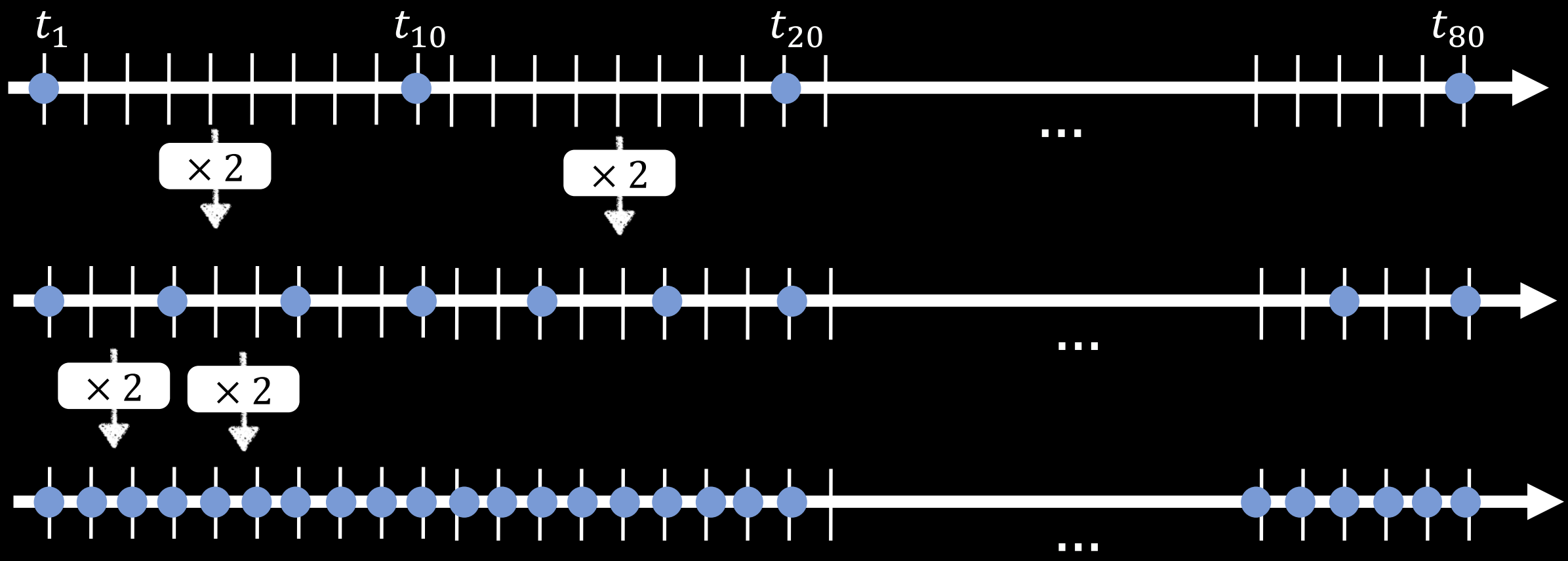


2) Fill-in missing frames



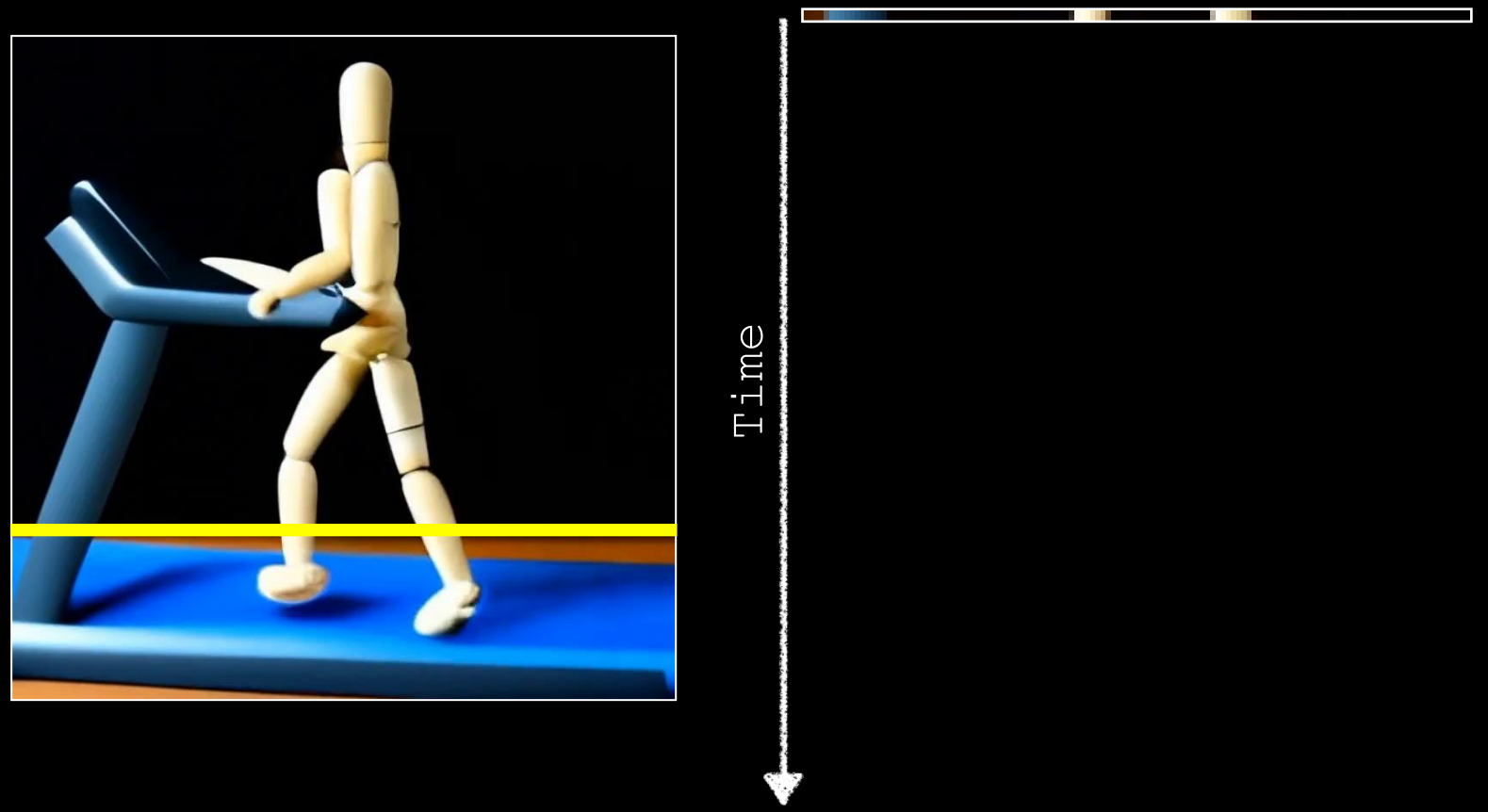
Common approach: distant keyframes → frame interpolation

In practice, requires a system of several frame interpolation models



Common approach: distant keyframes → frame interpolation

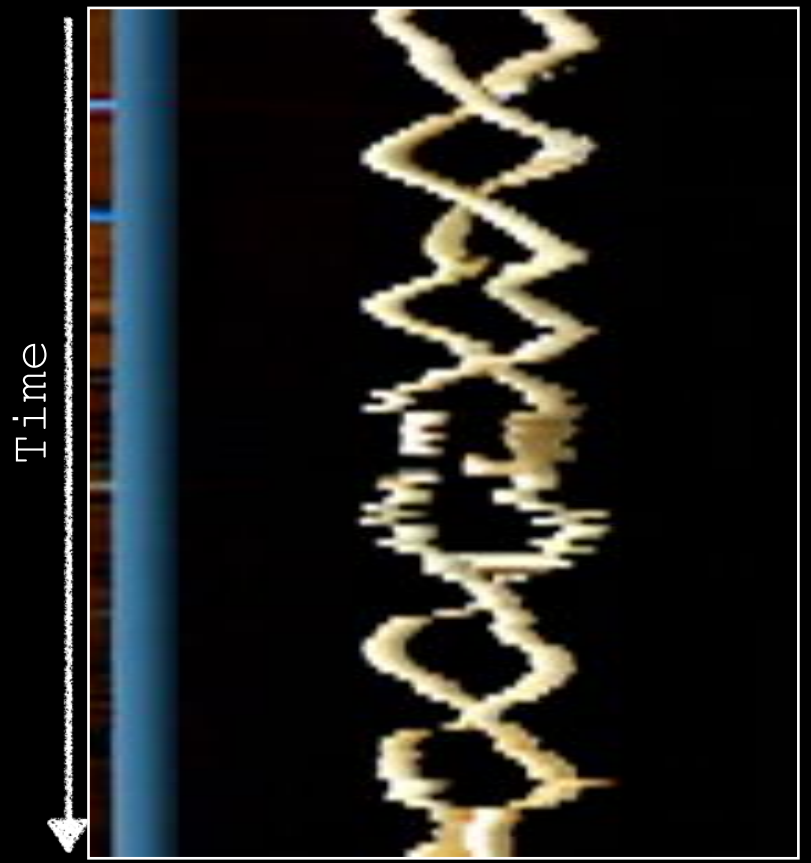
Aliasing ambiguities cannot be consistently resolved!



Common approach: distant keyframes → frame interpolation

Aliasing ambiguities cannot be consistently resolved!





Common approach: distant keyframes → frame interpolation

Mistakes in the keyframes are propagated in the interpolation stage



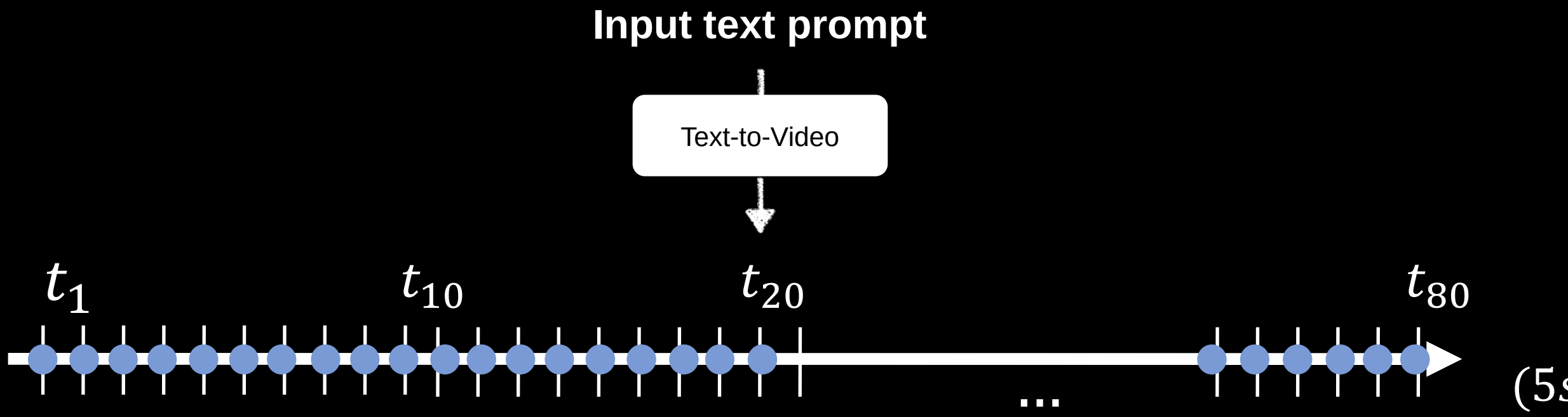








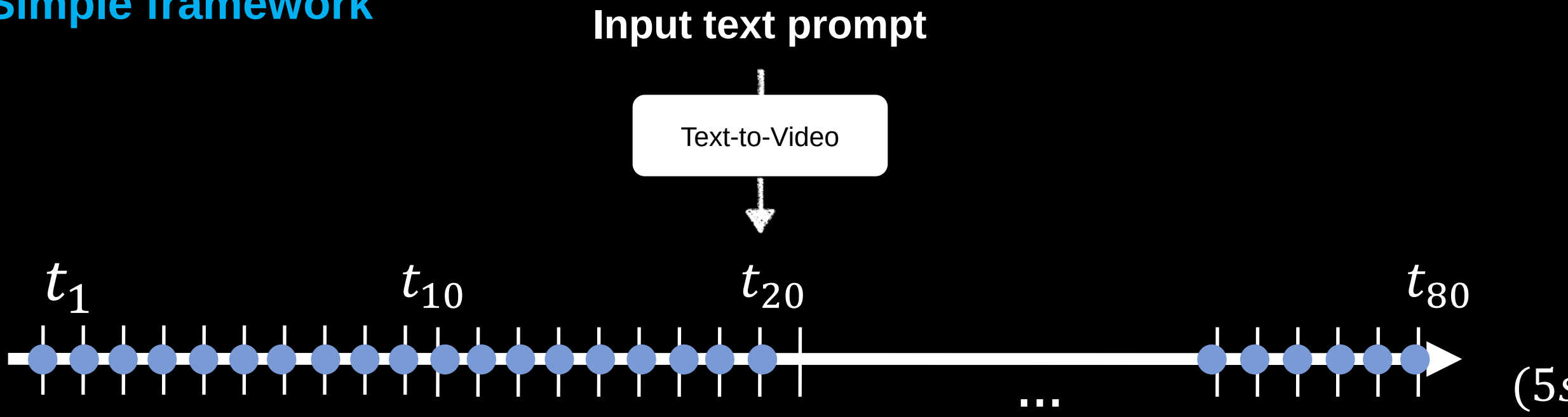
Our approach: generate all frames *at once*



Our approach: generate all frames *at once*

Allows learning globally coherent motion

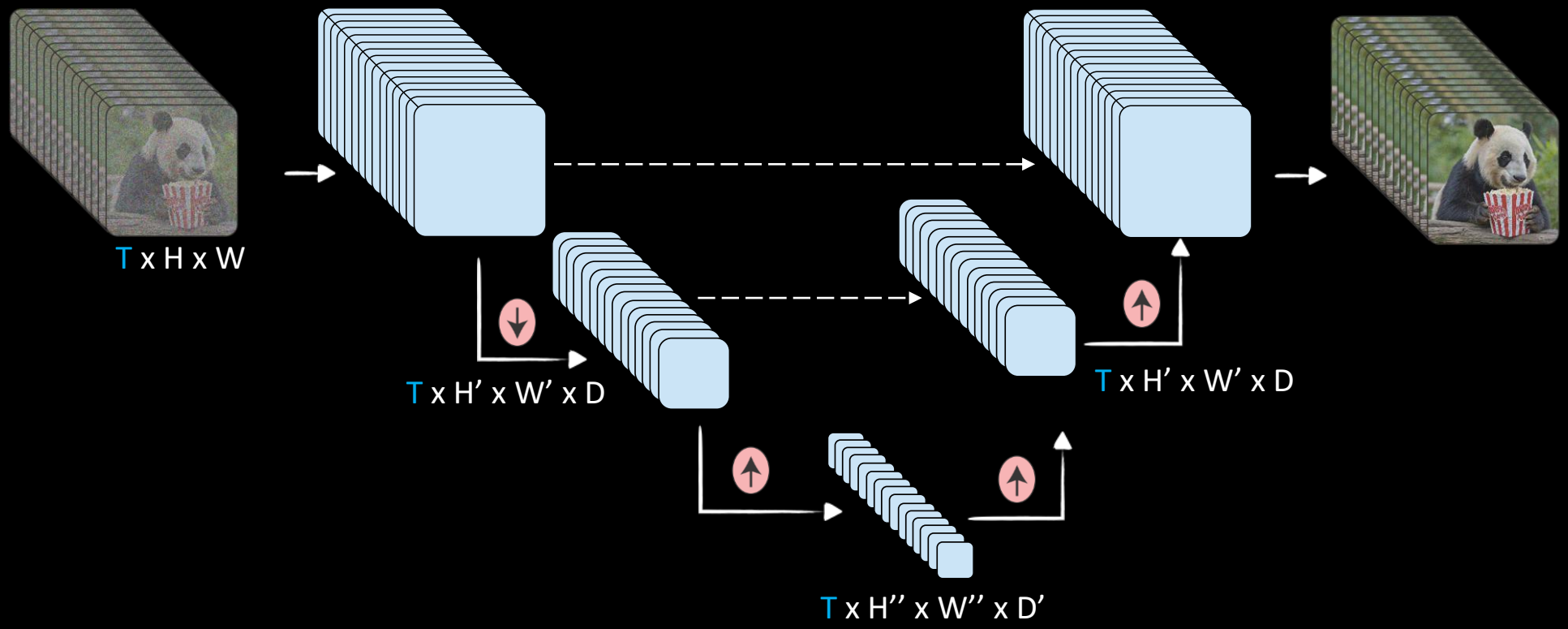
Simple framework



Common UNet

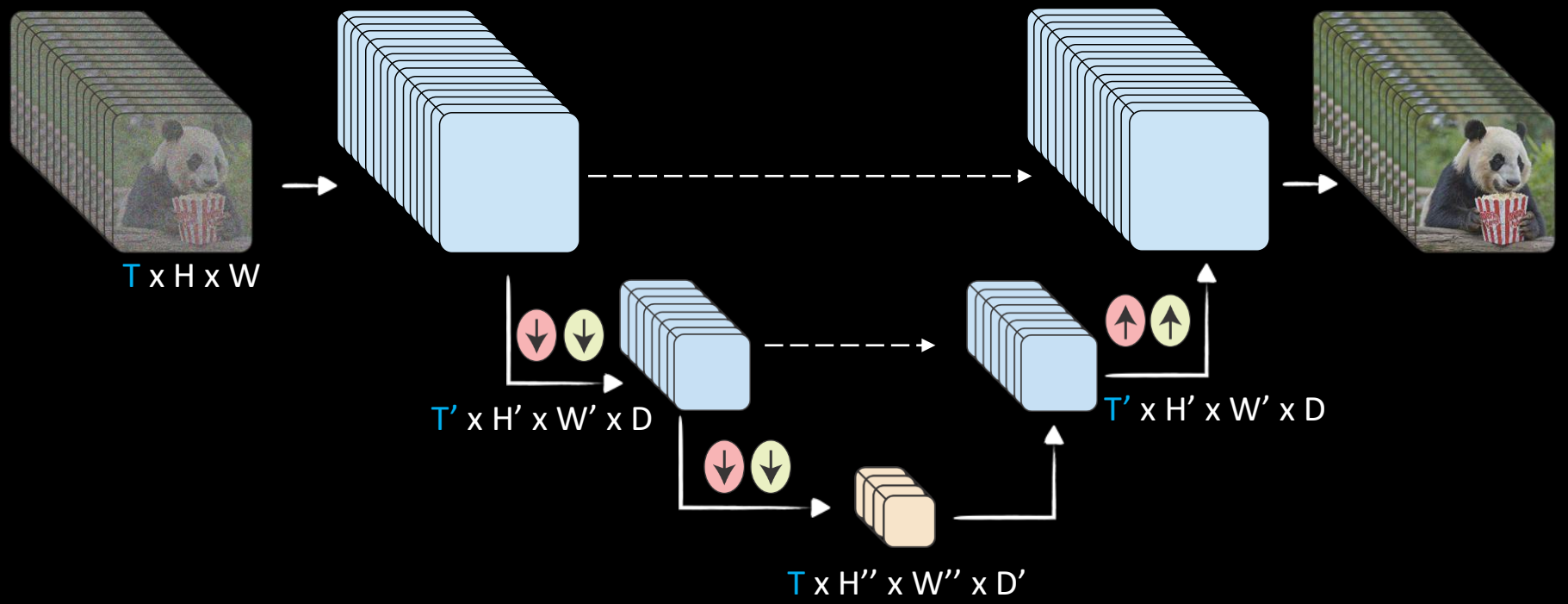
Spatial sub-sampling

No temporal sub-sampling



Space-Time UNet (STUNet)

Spatial sub-sampling
Temporal sub-sampling



Conv inflation block

- Pretrained T2I layer
- 2D Convolution
- Norm + activation
- 1D Convolution
- Norm + activation
- Linear Projection

Attention inflation block

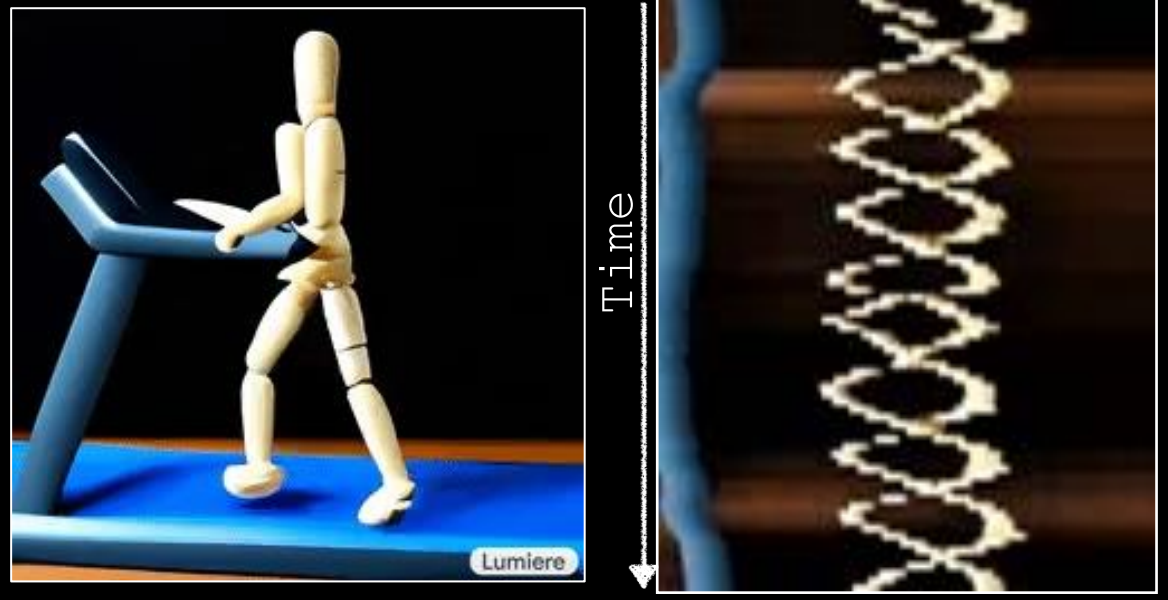
- Pretrained T2I layer
- 1D Attention (xL)
- Linear Projection

"3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation", Çiçek et al. *MICCAI 2016*

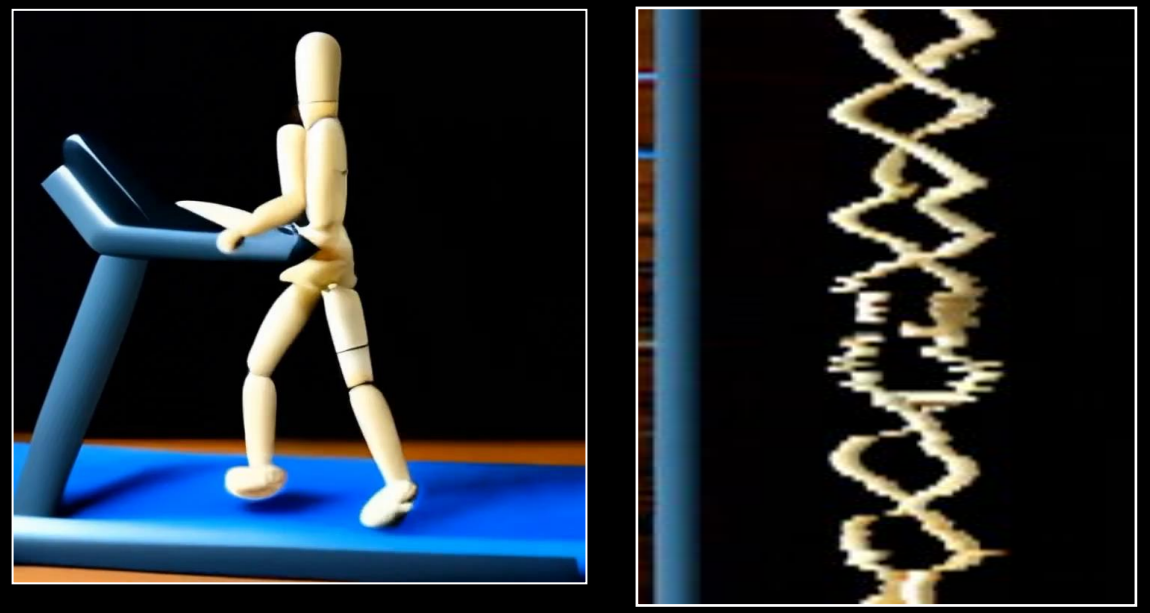
Space-Time UNet (STUNet)



Space-Time UNet (STUNet)



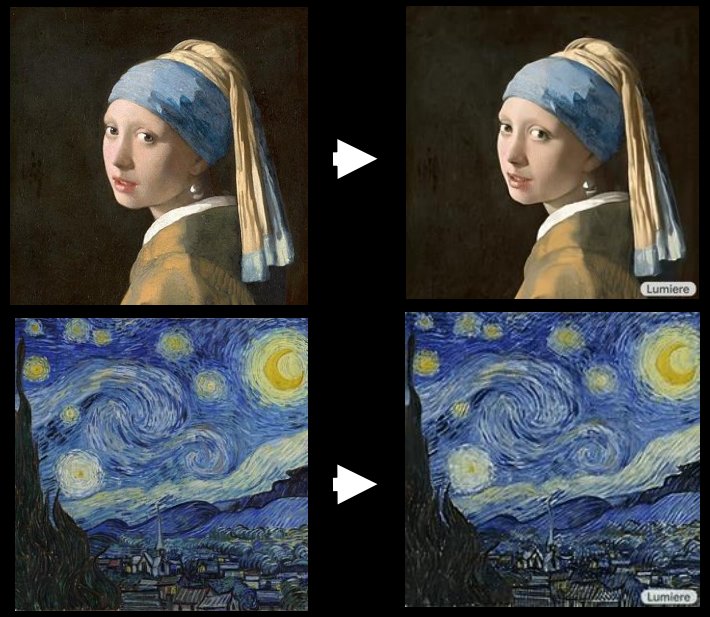
Cascaded Approach



Text-to-Video



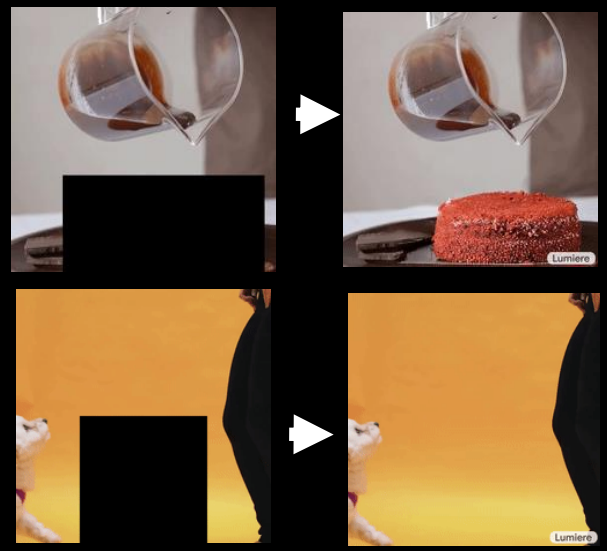
Image-to-Video



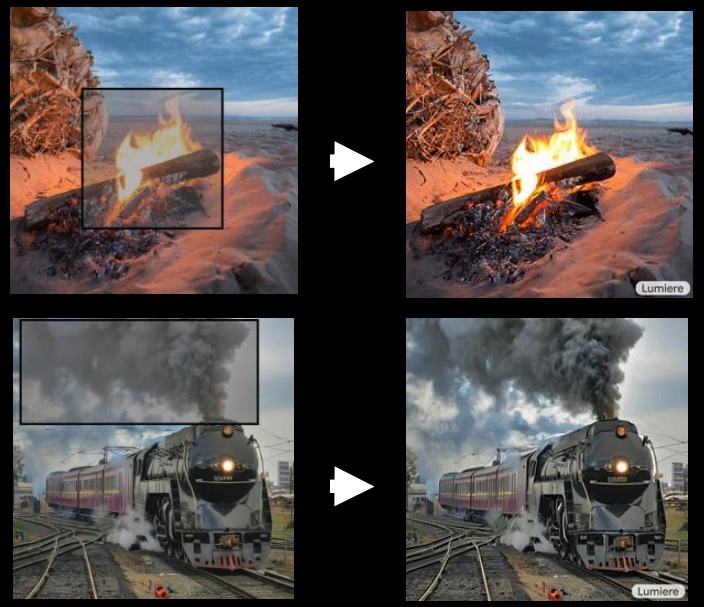
Stylized Generation



Inpainting



Cinemagraphs



Editing

